Corollary 18. Let n be a positive integer and let $p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ be its factorization into powers of distinct primes. Then

$$\mathbb{Z}/n\mathbb{Z} \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z}) \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z}) \times \cdots \times (\mathbb{Z}/p_k^{\alpha_k}\mathbb{Z}),$$

as rings, so in particular we have the following isomorphism of multiplicative groups:

$$(\mathbb{Z}/n\mathbb{Z})^\times \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z})^\times \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/p_k^{\alpha_k}\mathbb{Z})^\times.$$

If we compare orders on the two sides of this last isomorphism, we obtain the formula

$$\varphi(n) = \varphi(p_1^{\alpha_1})\varphi(p_2^{\alpha_2}) \dots \varphi(p_k^{\alpha_k})$$

for the Euler φ-function. This in turn implies that φ is what in elementary number theory is termed a *multiplicative function*, namely that $\varphi(ab) = \varphi(a)\varphi(b)$ whenever a and b are relatively prime positive integers. The value of φ on prime powers p^α is easily seen to be $\varphi(p^\alpha) = p^{\alpha-1}(p-1)$ (cf. Chapter 0). From this and the multiplicativity of φ we obtain its value on all positive integers.

Corollary 18 is also a step toward a determination of the decomposition of the abelian group $(\mathbb{Z}/n\mathbb{Z})^\times$ into a direct product of cyclic groups. The complete structure is derived at the end of Section 9.5.

EXERCISES

Let R be a ring with identity $1 \neq 0$.

1. An element $e \in R$ is called an *idempotent* if $e^2 = e$. Assume e is an idempotent in R and $er = re$ for all $r \in R$. Prove that Re and $R(1-e)$ are two-sided ideals of R and that $R \cong Re \times R(1-e)$. Show that e and $1-e$ are identities for the subrings Re and $R(1-e)$ respectively.
2. Let R be a finite Boolean ring with identity $1 \neq 0$ (cf. Exercise 15 of Section 1). Prove that $R \cong \mathbb{Z}/2\mathbb{Z} \times \cdots \times \mathbb{Z}/2\mathbb{Z}$. [Use the preceding exercise.]
3. Let R and S be rings with identities. Prove that every ideal of $R \times S$ is of the form $I \times J$ where I is an ideal of R and J is an ideal of S.
4. Prove that if R and S are nonzero rings then $R \times S$ is never a field.
5. Let $n_1, n_2, \dots, n_k$ be integers which are relatively prime in pairs: $(n_i, n_j) = 1$ for all $i \neq j$.
   (a) Show that the Chinese Remainder Theorem implies that for any $a_1, \dots, a_k \in \mathbb{Z}$ there is a solution $x \in \mathbb{Z}$ to the simultaneous congruences

   $$x \equiv a_1 \bmod n_1, \qquad x \equiv a_2 \bmod n_2, \qquad \dots, \qquad x \equiv a_k \bmod n_k$$

   and that the solution x is unique mod $n = n_1 n_2 \dots n_k$.
   (b) Let $n_i' = n/n_i$ be the quotient of n by n_i, which is relatively prime to n_i by assumption. Let t_i be the inverse of n_i' mod n_i. Prove that the solution x in (a) is given by

   $$x = a_1 t_1 n_1' + a_2 t_2 n_2' + \cdots + a_k t_k n_k' \bmod n.$$

   Note that the elements t_i can be quickly found by the Euclidean Algorithm as described in Section 2 of the Preliminaries chapter (writing $an_i + bn_i' = (n_i, n_i') = 1$ gives $t_i = b$) and that these then quickly give the solutions to the system of congruences above for any choice of $a_1, a_2, \dots, a_k$.

(c) Solve the simultaneous system of congruences

$$x \equiv 1 \bmod 8, \qquad x \equiv 2 \bmod 25, \quad \text{and} \quad x \equiv 3 \bmod 81$$

and the simultaneous system

$$y \equiv 5 \bmod 8, \qquad y \equiv 12 \bmod 25, \quad \text{and} \quad y \equiv 47 \bmod 81.$$

6. Let $f_1(x), f_2(x), \ldots, f_k(x)$ be polynomials with integer coefficients of the same degree d. Let $n_1, n_2, \ldots, n_k$ be integers which are relatively prime in pairs (i.e., $(n_i, n_j) = 1$ for all $i \neq j$). Use the Chinese Remainder Theorem to prove there exists a polynomial $f(x)$ with integer coefficients and of degree d with

$$f(x) \equiv f_1(x) \bmod n_1, \qquad f(x) \equiv f_2(x) \bmod n_2, \quad \ldots, \qquad f(x) \equiv f_k(x) \bmod n_k$$

i.e., the coefficients of $f(x)$ agree with the coefficients of $f_i(x) \bmod n_i$. Show that if all the $f_i(x)$ are monic, then $f(x)$ may also be chosen monic. [Apply the Chinese Remainder Theorem in $\mathbb{Z}$ to each of the coefficients separately.]

7. Let m and n be positive integers with n dividing m. Prove that the natural surjective ring projection $\mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ is also surjective on the units: $(\mathbb{Z}/m\mathbb{Z})^\times \to (\mathbb{Z}/n\mathbb{Z})^\times$.

The next four exercises develop the concept of *direct limits* and the "dual" notion of *inverse limits*. In these exercises I is a nonempty index set with a partial order $\leq$ (cf. Appendix I). For each $i \in I$ let A_i be an additive abelian group. In Exercise 8 assume also that I is a *directed set*: for every $i, j \in I$ there is some $k \in I$ with $i \leq k$ and $j \leq k$.

8. Suppose for every pair of indices i, j with $i \leq j$ there is a map $\rho_{ij} : A_i \to A_j$ such that the following hold:

i. $\rho_{jk} \circ \rho_{ij} = \rho_{ik}$ whenever $i \leq j \leq k$, and
ii. $\rho_{ii} = 1$ for all $i \in I$.

Let B be the disjoint union of all the A_i. Define a relation $\sim$ on B by

$$a \sim b \text{ if and only if there exists } k \text{ with } i, j \leq k \text{ and } \rho_{ik}(a) = \rho_{jk}(b),$$

for $a \in A_i$ and $b \in A_j$.

(a) Show that $\sim$ is an equivalence relation on B. (The set of equivalence classes is called the *direct* or *inductive limit* of the directed system $\{A_i\}$, and is denoted $\varinjlim A_i$. In the remaining parts of this exercise let $A = \varinjlim A_i$.)

(b) Let $\overline{x}$ denote the class of x in A and define $\rho_i : A_i \to A$ by $\rho_i(a) = \overline{a}$. Show that if each ρ_{ij} is injective, then so is ρ_i for all i (so we may then identify each A_i as a subset of A).

(c) Assume all ρ_{ij} are group homomorphisms. For $a \in A_i, b \in A_j$ show that the operation

$$\overline{a} + \overline{b} = \overline{\rho_{ik}(a) + \rho_{jk}(b)}$$

where k is any index with $i, j \leq k$, is well defined and makes A into an abelian group. Deduce that the maps ρ_i in (b) are group homomorphisms from A_i to A.

(d) Show that if all A_i are commutative rings with 1 and all ρ_{ij} are ring homomorphisms that send 1 to 1, then A may likewise be given the structure of a commutative ring with 1 such that all ρ_i are ring homomorphisms.

(e) Under the hypotheses in (c) prove that the direct limit has the following *universal property*: if C is any abelian group such that for each $i \in I$ there is a homomorphism $\varphi_i : A_i \to C$ with $\varphi_i = \varphi_j \circ \rho_{ij}$ whenever $i \leq j$, then there is a unique homomorphism $\varphi : A \to C$ such that $\varphi \circ \rho_i = \varphi_i$ for all i.

9. Let I be the collection of open intervals $U = (a, b)$ on the real line containing a fixed real number p. Order these by reverse inclusion: $U \le V$ if $V \subseteq U$ (note that I is a directed set). For each U let A_U be the ring of continuous real valued functions on U. For $V \subseteq U$ define the *restriction maps* $\rho_{UV} : A_U \to A_V$ by $f \mapsto f|_V$, the usual restriction of a function on U to a function on the subset V (which is easily seen to be a ring homomorphism). Let $A = \varinjlim A_U$ be the direct limit. In the notation of the preceding exercise, show that the maps $\rho_U : A_U \to A$ are *not* injective but are all surjective (A is called the ring of *germs of continuous functions* at p).

We now develop the notion of *inverse limits*. Continue to assume I is a partially ordered set (but not necessarily directed), and A_i is a group for all $i \in I$.

10. Suppose for every pair of indices i, j with $i \le j$ there is a map $\mu_{ji} : A_j \to A_i$ such that the following hold:

i. $\mu_{ji} \circ \mu_{kj} = \mu_{ki}$ whenever $i \le j \le k$, and
ii. $\mu_{ii} = 1$ for all $i \in I$.

Let P be the subset of elements $(a_i)_{i \in I}$ in the direct product $\prod_{i \in I} A_i$ such that $\mu_{ji}(a_j) = a_i$ whenever $i \le j$ (here a_i and a_j are the i^{th} and j^{th} components respectively of the element in the direct product). The set P is called the *inverse* or *projective limit* of the system $\{A_i\}$, and is denoted $\varprojlim A_i$.)

(a) Assume all μ_{ji} are group homomorphisms. Show that P is a subgroup of the direct product group (cf. Exercise 15, Section 5.1).
(b) Assume the hypotheses in (a), and let $I = \mathbb{Z}^+$ (usual ordering). For each $i \in I$ let $\mu_i : P \to A_i$ be the projection of P onto its i^{th} component. Show that if each μ_{ji} is surjective, then so is μ_i for all i (so each A_i is a quotient group of P).
(c) Show that if all A_i are commutative rings with 1 and all μ_{ji} are ring homomorphisms that send 1 to 1, then A may likewise be given the structure of a commutative ring with 1 such that all μ_i are ring homomorphisms.
(d) Under the hypotheses in (a) prove that the inverse limit has the following *universal property*: if D is any group such that for each $i \in I$ there is a homomorphism $\pi_i : D \to A_i$ with $\pi_i = \mu_{ji} \circ \pi_j$ whenever $i \le j$, then there is a unique homomorphism $\pi : D \to P$ such that $\mu_i \circ \pi = \pi_i$ for all i.

11. Let p be a prime let $I = \mathbb{Z}^+$, let $A_i = \mathbb{Z}/p^i\mathbb{Z}$ and let μ_{ji} be the natural projection maps

$$\mu_{ji} : a \pmod{p^j} \longmapsto a \pmod{p^i}.$$

The inverse limit $\varprojlim \mathbb{Z}/p^i\mathbb{Z}$ is called the ring of *p-adic integers*, and is denoted by $\mathbb{Z}_p$.

(a) Show that every element of $\mathbb{Z}_p$ may be written uniquely as an infinite formal sum $b_0 + b_1 p + b_2 p^2 + b_3 p^3 + \cdots$ with each $b_i \in \{0, 1, \dots, p-1\}$. Describe the rules for adding and multiplying such formal sums corresponding to addition and multiplication in the ring $\mathbb{Z}_p$. [Write a least residue in each $\mathbb{Z}/p^i\mathbb{Z}$ in its base p expansion and then describe the maps μ_{ji}.] (Note in particular that $\mathbb{Z}_p$ is uncountable.)
(b) Prove that $\mathbb{Z}_p$ is an integral domain that contains a copy of the integers.
(c) Prove that $b_0 + b_1 p + b_2 p^2 + b_3 p^3 + \cdots$ as in (a) is a unit in $\mathbb{Z}_p$ if and only if $b_0 \neq 0$.
(d) Prove that $p\mathbb{Z}_p$ is the unique maximal ideal of $\mathbb{Z}_p$ and $\mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{Z}/p\mathbb{Z}$ (where $p = 0 + 1p + 0p^2 + 0p^3 + \cdots$). Prove that every ideal of $\mathbb{Z}_p$ is of the form $p^n\mathbb{Z}_p$ for some integer $n \ge 0$.
(e) Show that if $a_1 \not\equiv 0 \pmod p$ then there is an element $a = (a_i)$ in the direct limit $\mathbb{Z}_p$ satisfying $a_j^p \equiv 1 \pmod{p^j}$ and $\mu_{j1}(a_j) = a_1$ for all j. Deduce that $\mathbb{Z}_p$ contains $p - 1$ distinct $(p-1)^{\text{st}}$ roots of 1.

CHAPTER 8

Euclidean Domains, Principal Ideal Domains, and Unique Factorization Domains

There are a number of classes of rings with more algebraic structure than generic rings. Those considered in this chapter are rings with a division algorithm (Euclidean Domains), rings in which every ideal is principal (Principal Ideal Domains) and rings in which elements have factorizations into primes (Unique Factorization Domains). The principal examples of such rings are the ring $\mathbb{Z}$ of integers and polynomial rings $F[x]$ with coefficients in some field F. We prove here all the theorems on the integers $\mathbb{Z}$ stated in the Preliminaries chapter as special cases of results valid for more general rings. These results will be applied to the special case of the ring $F[x]$ in the next chapter.

All rings in this chapter are commutative.

8.1 EUCLIDEAN DOMAINS

We first define the notion of a *norm* on an integral domain R. This is essentially no more than a measure of "size" in R.

Definition. Any function $N : R \to \mathbb{Z}^+ \cup \{0\}$ with $N(0) = 0$ is called a *norm* on the integral domain R. If $N(a) > 0$ for $a \neq 0$ define N to be a *positive norm.*

We observe that this notion of a norm is fairly weak and that it is possible for the same integral domain R to possess several different norms.

Definition. The integral domain R is said to be a *Euclidean Domain* (or possess a *Division Algorithm*) if there is a norm N on R such that for any two elements a and b of R with $b \neq 0$ there exist elements q and r in R with

$$a = qb + r \qquad \text{with } r = 0 \text{ or } N(r) < N(b).$$

The element q is called the *quotient* and the element r the *remainder* of the division.

The importance of the existence of a Division Algorithm on an integral domain R is that it allows a *Euclidean Algorithm* for two elements a and b of R: by successive "divisions" (these actually *are* divisions in the field of fractions of R) we can write

$$a = q_0 b + r_0 \tag{0}$$
$$b = q_1 r_0 + r_1 \tag{1}$$
$$r_0 = q_2 r_1 + r_2 \tag{2}$$
$$\vdots$$
$$r_{n-2} = q_n r_{n-1} + r_n \tag{n}$$
$$r_{n-1} = q_{n+1} r_n \tag{$n+1$}$$

where r_n is the last nonzero remainder. Such an r_n exists since $N(b) > N(r_0) > N(r_1) > \cdots > N(r_n)$ is a decreasing sequence of nonnegative integers if the remainders are nonzero, and such a sequence cannot continue indefinitely. Note also that there is no guarantee that these elements are *unique*.

Examples

(0) Fields are trivial examples of Euclidean Domains where any norm will satisfy the defining condition (e.g., $N(a) = 0$ for all a). This is because for every a, b with $b \neq 0$ we have $a = qb + 0$, where $q = ab^{-1}$.

(1) The integers $\mathbb{Z}$ are a Euclidean Domain with norm given by $N(a) = |a|$, the usual absolute value. The existence of a Division Algorithm in $\mathbb{Z}$ (the familiar "long division" of elementary arithmetic) is verified as follows. Let a and b be two nonzero integers and suppose first that $b > 0$. The half open intervals $[\,nb\,,\ (n{+}1)b\,)$, $n \in \mathbb{Z}$ partition the real line and so a is in one of them, say $a \in [\,kb\,,\ (k{+}1)b\,)$. For $q = k$ we have $a - qb = r \in [0, |b|)$ as needed. If $b < 0$ (so $-b > 0$), by what we have just seen there is an integer q such that $a = q(-b) + r$ with either $r = 0$ or $|r| < |-b|$; then $a = (-q)b + r$ satisfies the requirements of the Division Algorithm for a and b. This argument can be made more formal by using induction on $|a|$.

Note that if a is not a multiple of b there are always two possibilities for the pair q, r: the proof above always produced a positive remainder r. If for example $b > 0$ and q, r are as above with $r > 0$, then $a = q'b + r'$ with $q' = q + 1$ and $r' = r - b$ also satisfy the conditions of the Division Algorithm applied to a, b. Thus $5 = 2 \cdot 2 + 1 = 3 \cdot 2 - 1$ are the two ways of applying the Division Algorithm in $\mathbb{Z}$ to $a = 5$ and $b = 2$. The quotient and remainder are unique if we require the remainder to be nonnegative.

(2) If F is a field, then the polynomial ring $F[x]$ is a Euclidean Domain with norm given by $N(p(x)) =$ the degree of $p(x)$. The Division Algorithm for polynomials is simply "long division" of polynomials which may be familiar for polynomials with real coefficients. The proof is very similar to that for $\mathbb{Z}$ and is given in the next chapter (although for polynomials the quotient and remainder are shown to be unique). In order for a polynomial ring to be a Euclidean Domain the coefficients must come from a field since the division algorithm ultimately rests on being able to divide arbitrary nonzero coefficients. We shall prove in Section 2 that $R[x]$ is not a Euclidean Domain if R is not a field.

(3) The quadratic integer rings $\mathcal{O}$ in Section 7.1 are integral domains with a norm defined by the absolute value of the field norm (to ensure the values taken are nonnegative;

when $D < 0$ the field norm is itself a norm), but in general $\mathcal{O}$ is not Euclidean with respect to this norm (or any other norm). The Gaussian integers $\mathbb{Z}[i]$ (where $D = -1$), however, are a Euclidean Domain with respect to the norm $N(a+bi) = a^2 + b^2$, as we now show (cf. also the end of Section 3).

Let $\alpha = a+bi$, $\beta = c+di$ be two elements of $\mathbb{Z}[i]$ with $\beta \neq 0$. Then in the field $\mathbb{Q}(i)$ we have $\dfrac{\alpha}{\beta} = r+si$ where $r = (ac+bd)/(c^2+d^2)$ and $s = (bc-ad)/(c^2+d^2)$ are rational numbers. Let p be an integer closest to the rational number r and let q be an integer closest to the rational number s, so that both $|r-p|$ and $|s-q|$ are at most $1/2$. The Division Algorithm follows immediately once we show

$$\alpha = (p+qi)\beta + \gamma \qquad \text{for some } \gamma \in \mathbb{Z}[i] \text{ with } \quad N(\gamma) \leq \frac{1}{2}N(\beta)$$

which is even stronger than necessary. Let $\theta = (r-p) + (s-q)i$ and set $\gamma = \beta\theta$. Then $\gamma = \alpha - (p+qi)\beta$, so that $\gamma \in \mathbb{Z}[i]$ is a Gaussian integer and $\alpha = (p+qi)\beta + \gamma$. Since $N(\theta) = (r-p)^2 + (s-q)^2$ is at most $1/4 + 1/4 = 1/2$, the multiplicativity of the norm N implies that $N(\gamma) = N(\theta)N(\beta) \leq \frac{1}{2}N(\beta)$ as claimed.

Note that the algorithm is quite explicit since a quotient $p+qi$ is quickly determined from the rational numbers r and s, and then the remainder $\gamma = \alpha - (p+qi)\beta$ is easily computed. Note also that the quotient need not be unique: if r (or s) is half of an odd integer then there are two choices for p (or for q, respectively).

This proof that $\mathbb{Z}[i]$ is a Euclidean Domain can also be used to show that $\mathcal{O}$ is a Euclidean Domain (with respect to the field norm defined in Section 7.1) for $D = -2, -3, -7, -11$ (cf. the exercises). We shall see shortly that $\mathbb{Z}[\sqrt{-5}\,]$ is not Euclidean with respect to any norm, and a proof that $\mathbb{Z}[(1+\sqrt{-19}\,)/2]$ is not a Euclidean Domain with respect to any norm appears at the end of this section.

(4) Recall (cf. Exercise 26 in Section 7.1) that a *discrete valuation ring* is obtained as follows. Let K be a field. A *discrete valuation* on K is a function $\nu : K^\times \to \mathbb{Z}$ satisfying

(i) $\nu(ab) = \nu(a) + \nu(b)$ (i.e., ν is a homomorphism from the multiplicative group of nonzero elements of K to $\mathbb{Z}$),

(ii) ν is surjective, and

(iii) $\nu(x+y) \geq \min\{\nu(x), \nu(y)\}$ for all $x, y \in K^\times$ with $x + y \neq 0$.

The set $\{x \in K^\times \mid \nu(x) \geq 0\} \cup \{0\}$ is a subring of K called the valuation ring of ν. An integral domain R is called a discrete valuation ring if there is a valuation ν on its field of fractions such that R is the valuation ring of ν.

For example the ring R of all rational numbers whose denominators are relatively prime to the fixed prime $p \in \mathbb{Z}$ is a discrete valuation ring contained in $\mathbb{Q}$.

A discrete valuation ring is easily seen to be a Euclidean Domain with respect to the norm defined by $N(0) = 0$ and $N = \nu$ on the nonzero elements of R. This is because for $a, b \in R$ with $b \neq 0$

(a) if $N(a) < N(b)$ then $a = 0 \cdot b + a$, and

(b) if $N(a) \geq N(b)$ then it follows from property (i) of a discrete valuation that $q = ab^{-1} \in R$, so $a = qb + 0$.

The first implication of a Division Algorithm for the integral domain R is that it forces every ideal of R to be *principal*.

Proposition 1. Every ideal in a Euclidean Domain is principal. More precisely, if I is any nonzero ideal in the Euclidean Domain R then $I = (d)$, where d is any nonzero element of I of minimum norm.

Proof: If I is the zero ideal, there is nothing to prove. Otherwise let d be any nonzero element of I of minimum norm (such a d exists since the set $\{N(a) \mid a \in I\}$ has a minimum element by the Well Ordering of $\mathbb{Z}$). Clearly $(d) \subseteq I$ since d is an element of I. To show the reverse inclusion let a be any element of I and use the Division Algorithm to write $a = qd + r$ with $r = 0$ or $N(r) < N(d)$. Then $r = a - qd$ and both a and qd are in I, so r is also an element of I. By the minimality of the norm of d, we see that r must be 0. Thus $a = qd \in (d)$ showing $I = (d)$.

Proposition 1 shows that every ideal of $\mathbb{Z}$ is principal. This fundamental property of $\mathbb{Z}$ was previously determined (in Section 7.3) from the (additive) group structure of $\mathbb{Z}$, using the classification of the subgroups of cyclic groups in Section 2.3. Note that these are really the same proof, since the results in Section 2.3 ultimately relied on the Euclidean Algorithm in $\mathbb{Z}$.

Proposition 1 can also be used to prove that some integral domains R are *not* Euclidean Domains (with respect to *any* norm) by proving the existence of ideals of R that are not principal.

Examples

(1) Let $R = \mathbb{Z}[x]$. Since the ideal $(2, x)$ is not principal (cf. Example 3 at the beginning of Section 7.4), it follows that the ring $\mathbb{Z}[x]$ of polynomials with *integer* coefficients is *not* a Euclidean Domain (for any choice of norm), even though the ring $\mathbb{Q}[x]$ of polynomials with *rational* coefficients is a Euclidean Domain.

(2) Let R be the quadratic integer ring $\mathbb{Z}[\sqrt{-5}\,]$, let N be the associated field norm $N(a+b\sqrt{-5}\,) = a^2+5b^2$ and consider the ideal $I = (3, 2+\sqrt{-5}\,)$ generated by 3 and $2+\sqrt{-5}$. Suppose $I = (a+b\sqrt{-5}\,)$, $a, b \in \mathbb{Z}$, were principal, i.e., $3 = \alpha(a+b\sqrt{-5}\,)$ and $2+\sqrt{-5} = \beta(a+b\sqrt{-5}\,)$ for some $\alpha, \beta \in R$. Taking norms in the first equation gives $9 = N(\alpha)(a^2+5b^2)$ and since a^2+5b^2 is a positive integer it must be 1,3 or 9. If the value is 9 then $N(\alpha) = 1$ and $\alpha = \pm 1$, so $a+b\sqrt{-5} = \pm 3$, which is impossible by the second equation since the coefficients of $2+\sqrt{-5}$ are not divisible by 3. The value cannot be 3 since there are no integer solutions to $a^2+5b^2 = 3$. If the value is 1, then $a+b\sqrt{-5} = \pm 1$ and the ideal I would be the entire ring R. But then 1 would be an element of I, so $3\gamma + (2+\sqrt{-5}\,)\delta = 1$ for some $\gamma, \delta \in R$. Multiplying both sides by $2-\sqrt{-5}$ would then imply that $2-\sqrt{-5}$ is a multiple of 3 in R, a contradiction. It follows that I is not a principal ideal and so R is not a Euclidean Domain (with respect to any norm).

One of the fundamental consequences of the Euclidean Algorithm in $\mathbb{Z}$ is that it produces a greatest common divisor of two nonzero elements. This is true in any Euclidean Domain. The notion of a greatest common divisor of two elements (if it exists) can be made precise in general rings.

Definition. Let R be a commutative ring and let $a, b \in R$ with $b \neq 0$.

(1) a is said to be a *multiple* of b if there exists an element $x \in R$ with $a = bx$. In this case b is said to *divide* a or be a *divisor* of a, written $b \mid a$.

(2) A *greatest common divisor* of a and b is a nonzero element d such that

  (i) $d \mid a$ and $d \mid b$, and

  (ii) if $d' \mid a$ and $d' \mid b$ then $d' \mid d$.

A greatest common divisor of a and b will be denoted by g.c.d.(a, b), or (abusing the notation) simply (a, b).

Note that $b \mid a$ in a ring R if and only if $a \in (b)$ if and only if $(a) \subseteq (b)$. In particular, if d is any divisor of both a and b then (d) must contain both a and b and hence must contain the ideal generated by a and b. The defining properties (i) and (ii) of a greatest common divisor of a and b translated into the language of ideals therefore become (respectively):

if I is the ideal of R generated by a and b, then d is a greatest common divisor of a and b if

(i) I is contained in the principal ideal (d), and

(ii) if (d') is any principal ideal containing I then $(d) \subseteq (d')$.

Thus a greatest common divisor of a and b (if such exists) is a generator for the unique smallest principal ideal containing a and b. There are rings in which greatest common divisors do not exist.

This discussion immediately gives the following *sufficient* condition for the existence of a greatest common divisor.

Proposition 2. If a and b are nonzero elements in the commutative ring R such that the ideal generated by a and b is a principal ideal (d), then d is a greatest common divisor of a and b.

This explains why the symbol (a, b) is often used to denote both the ideal generated by a and b and a greatest common divisor of a and b. An integral domain in which every ideal (a, b) generated by two elements is principal is called a *Bezout Domain.* The exercises in this and subsequent sections explore these rings and show that there are Bezout Domains containing nonprincipal (necessarily infinitely generated) ideals.

Note that the condition in Proposition 2 is *not* a *necessary* condition. For example, in the ring $R = \mathbb{Z}[x]$ the elements 2 and x generate a maximal, nonprincipal ideal (cf. the examples in Section 7.4). Thus $R = (1)$ is the unique principal ideal containing both 2 and x, so 1 is a greatest common divisor of 2 and x. We shall see other examples along these lines in Section 3.

Before returning to Euclidean Domains we examine the uniqueness of greatest common divisors.

Proposition 3. Let R be an integral domain. If two elements d and d' of R generate the same principal ideal, i.e., $(d) = (d')$, then $d' = ud$ for some unit u in R. In particular, if d and d' are both greatest common divisors of a and b, then $d' = ud$ for some unit u.

Proof: This is clear if either d or d' is zero so we may assume d and d' are nonzero. Since $d \in (d')$ there is some $x \in R$ such that $d = xd'$. Since $d' \in (d)$ there is some $y \in R$ such that $d' = yd$. Thus $d = xyd$ and so $d(1 - xy) = 0$. Since $d \neq 0$, $xy = 1$, that is, both x and y are units. This proves the first assertion. The second assertion follows from the first since any two greatest common divisors of a and b generate the same principal ideal (they divide each other).

One of the most important properties of Euclidean Domains is that greatest common divisors always exist and *can be computed algorithmically.*

Theorem 4. Let R be a Euclidean Domain and let a and b be nonzero elements of R. Let $d = r_n$ be the last nonzero remainder in the Euclidean Algorithm for a and b described at the beginning of this chapter. Then

(1) d is a greatest common divisor of a and b, and

(2) the principal ideal (d) is the ideal generated by a and b. In particular, d can be written as an *R-linear combination* of a and b, i.e., there are elements x and y in R such that

$$d = ax + by.$$

Proof: By Proposition 1, the ideal generated by a and b is principal so a, b do have a greatest common divisor, namely any element which generates the (principal) ideal (a, b). Both parts of the theorem will follow therefore once we show $d = r_n$ generates this ideal, i.e., once we show that

(i) $d \mid a$ and $d \mid b$ (so $(a, b) \subseteq (d)$)

(ii) d is an R-linear combination of a and b (so $(d) \subseteq (a, b)$).

To prove that d divides both a and b simply keep track of the divisibilities in the Euclidean Algorithm. Starting from the $(n+1)^{\text{st}}$ equation, $r_{n-1} = q_{n+1}r_n$, we see that $r_n \mid r_{n-1}$. Clearly $r_n \mid r_n$. By induction (proceeding from index n downwards to index 0) assume r_n divides r_{k+1} and r_k. By the $(k+1)^{\text{st}}$ equation, $r_{k-1} = q_{k+1}r_k + r_{k+1}$, and since r_n divides both terms on the right hand side we see that r_n also divides r_{k-1}. From the 1^{st} equation in the Euclidean Algorithm we obtain that r_n divides b and then from the 0^{th} equation we get that r_n divides a. Thus (i) holds.

To prove that r_n is in the ideal (a, b) generated by a and b proceed similarly by induction proceeding from equation (0) to equation (n). It follows from equation (0) that $r_0 \in (a, b)$ and by equation (1) that $r_1 = b - q_1 r_0 \in (b, r_0) \subseteq (a, b)$. By induction assume $r_{k-1}, r_k \in (a, b)$. Then by the $(k+1)^{\text{st}}$ equation

$$r_{k+1} = r_{k-1} - q_{k+1}r_k \in (r_{k-1}, r_k) \subseteq (a, b).$$

This induction shows $r_n \in (a, b)$, which completes the proof.

Much of the material above may be familiar from elementary arithmetic in the case of the integers $\mathbb{Z}$, except possibly for the translation into the language of ideals. For example, if $a = 2210$ and $b = 1131$ then the smallest ideal of $\mathbb{Z}$ that contains both a and b (the ideal generated by a and b) is $13\mathbb{Z}$, since 13 is the greatest common divisor of 2210 and 1131. This fact follows quickly from the Euclidean Algorithm:

$$2210 = 1 \cdot 1131 + 1079$$

$$1131 = 1 \cdot 1079 + 52$$
$$1079 = 20 \cdot 52 + 39$$
$$52 = 1 \cdot 39 + 13$$
$$39 = 3 \cdot 13$$

so that $13 = (2210, 1131)$ is the last nonzero remainder. Using the procedure of Theorem 4 we can also write 13 as a linear combination of 2210 and 1131 by first solving the next to last equation above for $13 = 52 - 1 \cdot 39$, then using previous equations to solve for 39 and 52, etc., finally writing 13 entirely in terms of 2210 and 1131. The answer in this case is

$$13 = (-22) \cdot 2210 + 43 \cdot 1131.$$

The Euclidean Algorithm in the integers $\mathbb{Z}$ is extremely fast. It is a theorem that the number of steps required to determine the greatest common divisor of two integers a and b is at worst 5 times the number of digits of the smaller of the two numbers. Put another way, this algorithm is *logarithmic* in the size of the integers. To obtain an appreciation of the speed implied here, notice that for the example above we would have expected at worst $5 \cdot 4 = 20$ divisions (the example required far fewer). If we had started with integers on the order of 10^{100} (large numbers by physical standards), we would have expected at worst only 500 divisions.

There is no uniqueness statement for the integers x and y in $(a, b) = ax + by$. Indeed, $x' = x + b$ and $y' = y - a$ satisfy $(a, b) = ax' + by'$. This is essentially the only possibility — one can prove that if x_0 and y_0 are solutions to the equation $ax + by = N$, then any other solutions x and y to this equation are of the form

$$x = x_0 + m\frac{b}{(a, b)}$$
$$y = y_0 - m\frac{a}{(a, b)}$$

for some integer m (positive or negative).

This latter theorem (a proof of which is outlined in the exercises) provides a complete solution of the *First Order Diophantine Equation* $ax + by = N$ provided we know there is *at least one* solution to this equation. But the equation $ax + by = N$ is simply another way of stating that N is an element of the ideal generated by a and b. Since we know this ideal is just (d), the principal ideal generated by the greatest common divisor d of a and b, this is the same as saying $N \in (d)$, i.e., N is divisible by d. Hence, *the equation $ax + by = N$ is solvable in integers x and y if and only if N is divisible by the g.c.d. of a and b* (and then the result quoted above gives a full set of solutions of this equation).

We end this section with another criterion that can sometimes be used to prove that a given integral domain is not a Euclidean Domain.[1] For any integral domain let

[1]The material here and in some of the following section follows the exposition by J.C. Wilson in *A principal ideal ring that is not a Euclidean ring*, Math. Mag., 46(1973), pp. 34–38, of ideas of Th. Motzkin, and use a simplification by Kenneth S. Williams in *Note on non-Euclidean Principal Ideal Domains*, Math. Mag., 48(1975), pp. 176–177.

$\widetilde{R} = R^\times \cup \{0\}$ denote the collection of units of R together with 0. An element $u \in R - \widetilde{R}$ is called a *universal side divisor* if for every $x \in R$ there is some $z \in \widetilde{R}$ such that u divides $x - z$ in R, i.e., there is a type of "division algorithm" for u: every x may be written $x = qu + z$ where z is either zero or a unit. The existence of universal side divisors is a weakening of the Euclidean condition:

Proposition 5. Let R be an integral domain that is not a field. If R is a Euclidean Domain then there are universal side divisors in R.

Proof: Suppose R is Euclidean with respect to some norm N and let u be an element of $R - \widetilde{R}$ (which is nonempty since R is not a field) of minimal norm. For any $x \in R$, write $x = qu + r$ where r is either 0 or $N(r) < N(u)$. In either case the minimality of u implies $r \in \widetilde{R}$. Hence u is a universal side divisor in R.

Example

We can use Proposition 5 to prove that the quadratic integer ring $R = \mathbb{Z}[(1+\sqrt{-19})/2]$ is not a Euclidean Domain with respect to any norm by showing that R contains no universal side divisors (we shall see in the next section that all of the ideals in R are principal, so the technique in the examples following Proposition 1 do not apply to this ring). We have already determined that ± 1 are the only units in R and so $\widetilde{R} = \{0, \pm 1\}$. Suppose $u \in R$ is a universal side divisor and let $N(a + b(1 + \sqrt{-19})/2) = a^2 + ab + 5b^2$ denote the field norm on R as in Section 7.1. Note that if $a, b \in \mathbb{Z}$ and $b \neq 0$ then $a^2 + ab + 5b^2 = (a + b/2)^2 + 19/4b^2 \geq 5$ and so the smallest nonzero values of N on R are 1 (for the units ± 1) and 4 (for ± 2). Taking $x = 2$ in the definition of a universal side divisor it follows that u must divide one of $2 - 0$ or 2 ± 1 in R, i.e., u is a nonunit divisor of 2 or 3 in R. If $2 = \alpha\beta$ then $4 = N(\alpha)N(\beta)$ and by the remark above it follows that one of α or β has norm 1, i.e., equals ± 1. Hence the only divisors of 2 in R are $\{\pm 1, \pm 2\}$. Similarly, the only divisors of 3 in R are $\{\pm 1, \pm 3\}$, so the only possible values for u are ± 2 or ± 3. Taking $x = (1+\sqrt{-19})/2$ it is easy to check that none of $x, x \pm 1$ are divisible by ± 2 or ± 3 in R, so none of these is a universal side divisor.

EXERCISES

1. For each of the following five pairs of integers a and b, determine their greatest common divisor d and write d as a linear combination $ax + by$ of a and b.
   (a) $a = 20, b = 13$.
   (b) $a = 69, b = 372$.
   (c) $a = 11391, b = 5673$.
   (d) $a = 507885, b = 60808$.
   (e) $a = 91442056588823$, $b = 779086434385541$ (the Euclidean Algorithm requires only 7 steps for these integers).
2. For each of the following pairs of integers a and n, show that a is relatively prime to n and determine the inverse of a mod n (cf. Section 3 of the Preliminaries chapter).
   (a) $a = 13, n = 20$.
   (b) $a = 69, n = 89$.
   (c) $a = 1891, n = 3797$.

(d) $a = 6003722857$, $n = 77695236973$ (the Euclidean Algorithm requires only 3 steps for these integers).

3. Let R be a Euclidean Domain. Let m be the minimum integer in the set of norms of nonzero elements of R. Prove that every nonzero element of R of norm m is a unit. Deduce that a nonzero element of norm zero (if such an element exists) is a unit.

4. Let R be a Euclidean Domain.
   (a) Prove that if $(a, b) = 1$ and a divides bc, then a divides c. More generally, show that if a divides bc with nonzero a, b then $\dfrac{a}{(a, b)}$ divides c.
   (b) Consider the Diophantine Equation $ax + by = N$ where a, b and N are integers and a, b are nonzero. Suppose x_0, y_0 is a solution: $ax_0 + by_0 = N$. Prove that the full set of solutions to this equation is given by
   $$x = x_0 + m\frac{b}{(a, b)}, \qquad y = y_0 - m\frac{a}{(a, b)}$$
   as m ranges over the integers. [If x, y is a solution to $ax + by = N$, show that $a(x - x_0) = b(y_0 - y)$ and use (a).]

5. Determine all integer solutions of the following equations:
   (a) $2x + 4y = 5$
   (b) $17x + 29y = 31$
   (c) $85x + 145y = 505$.

6. (*The Postage Stamp Problem*) Let a and b be two relatively prime positive integers. Prove that every sufficiently large positive integer N can be written as a linear combination $ax + by$ of a and b where x and y are both *nonnegative*, i.e., there is an integer N_0 such that for all $N \geq N_0$ the equation $ax + by = N$ can be solved with both x and y nonnegative integers. Prove in fact that the integer $ab - a - b$ cannot be written as a positive linear combination of a and b but that every integer greater than $ab - a - b$ is a positive linear combination of a and b (so every "postage" greater than $ab - a - b$ can be obtained using only stamps in denominations a and b).

7. Find a generator for the ideal $(85, 1+13i)$ in $\mathbb{Z}[i]$, i.e., a greatest common divisor for 85 and $1+13i$, by the Euclidean Algorithm. Do the same for the ideal $(47 - 13i, 53 + 56i)$.

It is known (but not so easy to prove) that $D = -1, -2, -3, -7, -11, -19, -43, -67$, and -163 are the only negative values of D for which every ideal in $\mathcal{O}$ is principal (i.e., $\mathcal{O}$ is a P.I.D. in the terminology of the next section). The results of the next exercise determine precisely which quadratic integer rings with $D < 0$ are Euclidean.

8. Let $F = \mathbb{Q}(\sqrt{D})$ be a quadratic field with associated quadratic integer ring $\mathcal{O}$ and field norm N as in Section 7.1.
   (a) Suppose D is $-1, -2, -3, -7$ or -11. Prove that $\mathcal{O}$ is a Euclidean Domain with respect to N. [Modify the proof for $\mathbb{Z}[i]$ ($D = -1$) in the text. For $D = -3, -7, -11$ prove that every element of F differs from an element in $\mathcal{O}$ by an element whose norm is at most $(1 + |D|)^2/(16|D|)$, which is less than 1 for these values of D. Plotting the points of $\mathcal{O}$ in $\mathbb{C}$ may be helpful.]
   (b) Suppose that $D = -43, -67$, or -163. Prove that $\mathcal{O}$ is not a Euclidean Domain with respect to any norm. [Apply the same proof as for $D = -19$ in the text.]

9. Prove that the ring of integers $\mathcal{O}$ in the quadratic integer ring $\mathbb{Q}(\sqrt{2})$ is a Euclidean Domain with respect to the norm given by the absolute value of the field norm N in Section 7.1.

10. Prove that the quotient ring $\mathbb{Z}[i]/I$ is finite for any nonzero ideal I of $\mathbb{Z}[i]$. [Use the fact

that $I = (\alpha)$ for some nonzero α and then use the Division Algorithm in this Euclidean Domain to see that every coset of I is represented by an element of norm less than $N(\alpha)$.]

11. Let R be a commutative ring with 1 and let a and b be nonzero elements of R. A *least common multiple* of a and b is an element e of R such that

(i) $a \mid e$ and $b \mid e$, and

(ii) if $a \mid e'$ and $b \mid e'$ then $e \mid e'$.

(a) Prove that a least common multiple of a and b (if such exists) is a generator for the unique largest principal ideal contained in $(a) \cap (b)$.

(b) Deduce that any two nonzero elements in a Euclidean Domain have a least common multiple which is unique up to multiplication by a unit.

(c) Prove that in a Euclidean Domain the least common multiple of a and b is $\dfrac{ab}{(a, b)}$, where (a, b) is the greatest common divisor of a and b.

12. (*A Public Key Code*) Let N be a positive integer. Let M be an integer relatively prime to N and let d be an integer relatively prime to $\varphi(N)$, where φ denotes Euler's φ-function. Prove that if $M_1 \equiv M^d \pmod N$ then $M \equiv M_1^{d'} \pmod N$ where d' is the inverse of d mod $\varphi(N)$: $dd' \equiv 1 \pmod{\varphi(N)}$.

Remark: This result is the basis for a standard *Public Key Code*. Suppose $N = pq$ is the product of two distinct large primes (each on the order of 100 digits, for example). If M is a message, then $M_1 \equiv M^d \pmod N$ is a scrambled (*encoded*) version of M, which can be unscrambled (*decoded*) by computing $M_1^{d'} \pmod N$ (these powers can be computed quite easily even for large values of M and N by successive squarings). The values of N and d (but not p and q) are made publicly known (hence the name) and then anyone with a message M can send their encoded message $M^d \pmod N$. To decode the message it seems necessary to determine d', which requires the determination of the value $\varphi(N) = \varphi(pq) = (p - 1)(q - 1)$ (no one has as yet *proved* that there is no other decoding scheme, however). The success of this method as a code rests on the necessity of determining the *factorization* of N into primes, for which no sufficiently efficient algorithm exists (for example, the most naive method of checking all factors up to $\sqrt{N}$ would here require on the order of 10^{100} computations, or approximately 300 years even at 10 billion computations per second, and of course one can always increase the size of p and q).

8.2 PRINCIPAL IDEAL DOMAINS (P.I.D.s)

Definition. A *Principal Ideal Domain* (P.I.D.) is an integral domain in which every ideal is principal.

Proposition 1 proved that *every Euclidean Domain is a Principal Ideal Domain* so that every result about Principal Ideal Domains automatically holds for Euclidean Domains.

Examples

(1) As mentioned after Proposition 1, the integers $\mathbb{Z}$ are a P.I.D. We saw in Section 7.4 that the polynomial ring $\mathbb{Z}[x]$ contains nonprincipal ideals, hence is not a P.I.D.

(2) Example 2 following Proposition 1 showed that the quadratic integer ring $\mathbb{Z}[\sqrt{-5}]$ is not a P.I.D., in fact the ideal $(3, 1 + \sqrt{-5})$ is a nonprincipal ideal. It is possible for the product IJ of two nonprincipal ideals I and J to be principal, for example the ideals $(3, 1+\sqrt{-5})$ and $(3, 1-\sqrt{-5})$ are both nonprincipal and their product is the principal ideal generated by 3, i.e., $(3, 1+\sqrt{-5})(3, 1-\sqrt{-5}) = (3)$ (cf. Exercise 5 and the example preceding Proposition 12 below).

It is not true that every Principal Ideal Domain is a Euclidean Domain. We shall prove below that the quadratic integer ring $\mathbb{Z}[(1+\sqrt{-19})/2]$, which was shown not to be a Euclidean Domain in the previous section, nevertheless is a P.I.D.

From an ideal-theoretic point of view Principal Ideal Domains are a natural class of rings to study beyond rings which are fields (where the ideals are just the trivial ones: (0) and (1)). Many of the properties enjoyed by Euclidean Domains are also satisfied by Principal Ideal Domains. A significant advantage of Euclidean Domains over Principal Ideal Domains, however, is that although greatest common divisors exist in both settings, in Euclidean Domains one has an *algorithm* for computing them. Thus (as we shall see in Chapter 12 in particular) results which depend on the existence of greatest common divisors may often be proved in the larger class of Principal Ideal Domains although computation of examples (i.e., concrete applications of these results) are more effectively carried out using a Euclidean Algorithm (if one is available).

We collect some facts about greatest common divisors proved in the preceding section.

Proposition 6. Let R be a Principal Ideal Domain and let a and b be nonzero elements of R. Let d be a generator for the principal ideal generated by a and b. Then

(1) d is a greatest common divisor of a and b

(2) d can be written as an *R-linear combination* of a and b, i.e., there are elements x and y in R with
$$d = ax + by$$

(3) d is unique up to multiplication by a unit of R.

Proof: This is just Propositions 2 and 3.

Recall that maximal ideals are always prime ideals but the converse is not true in general. We observed in Section 7.4, however, that every nonzero prime ideal of $\mathbb{Z}$ is a maximal ideal. This useful fact is true in an arbitrary Principal Ideal Domain, as the following proposition shows.

Proposition 7. Every nonzero prime ideal in a Principal Ideal Domain is a maximal ideal.

Proof: Let (p) be a nonzero prime ideal in the Principal Ideal Domain R and let $I = (m)$ be any ideal containing (p). We must show that $I = (p)$ or $I = R$. Now $p \in (m)$ so $p = rm$ for some $r \in R$. Since (p) is a prime ideal and $rm \in (p)$, either r or m must lie in (p). If $m \in (p)$ then $(p) = (m) = I$. If, on the other hand, $r \in (p)$ write $r = ps$. In this case $p = rm = psm$, so $sm = 1$ (recall that R is an integral domain) and m is a unit so $I = R$.

As we have already mentioned, if F is a field, then the polynomial ring $F[x]$ is a Euclidean Domain, hence also a Principal Ideal Domain (this will be proved in the next chapter). The converse to this is also true. Intuitively, if I is an ideal in R (such as the ideal (2) in $\mathbb{Z}$) then the ideal (I, x) in $R[x]$ (such as the ideal $(2, x)$ in $\mathbb{Z}[x]$) requires one more generator than does I, hence in general is not principal.

Corollary 8. If R is any commutative ring such that the polynomial ring $R[x]$ is a Principal Ideal Domain (or a Euclidean Domain), then R is necessarily a field.

Proof: Assume $R[x]$ is a Principal Ideal Domain. Since R is a subring of $R[x]$ then R must be an integral domain (recall that $R[x]$ has an identity if and only if R does). The ideal (x) is a nonzero prime ideal in $R[x]$ because $R[x]/(x)$ is isomorphic to the integral domain R. By Proposition 7, (x) is a maximal ideal, hence the quotient R is a field by Proposition 12 in Section 7.4.

The last result in this section will be used to prove that not every P.I.D. is a Euclidean Domain and relates the principal ideal property with another weakening of the Euclidean condition.

Definition. Define N to be a *Dedekind–Hasse norm* if N is a positive norm and for every nonzero $a, b \in R$ either a is an element of the ideal (b) or there is a nonzero element in the ideal (a, b) of norm strictly smaller than the norm of b (i.e., either b divides a in R or there exist $s, t \in R$ with $0 < N(sa - tb) < N(b)$).

Note that R is Euclidean with respect to a positive norm N if it is always possible to satisfy the Dedekind–Hasse condition with $s = 1$, so this is indeed a weakening of the Euclidean condition.

Proposition 9. The integral domain R is a P.I.D. if and only if R has a Dedekind–Hasse norm.[2]

Proof: Let I be any nonzero ideal in R and let b be a nonzero element of I with $N(b)$ minimal. Suppose a is any nonzero element in I, so that the ideal (a, b) is contained in I. Then the Dedekind–Hasse condition on N and the minimality of b implies that $a \in (b)$, so $I = (b)$ is principal. The converse will be proved in the next section (Corollary 16).

[2]That a Dedekind–Hasse norm on R implies that R is a P.I.D. (and is equivalent when R is a ring of algebraic integers) is the classical *Criterion of Dedekind and Hasse*, cf. *Über eindeutige Zerlegung in Primelemente oder in Primhauptideale in Integritätsbereichen*, Jour. für die Reine und Angew. Math., 159(1928), pp. 3–12. The observation that the converse holds generally is more recent and due to John Greene, *Principal Ideal Domains are almost Euclidean*, Amer. Math. Monthly, 104(1997), pp. 154–156.

Example

Let $R = \mathbb{Z}[(1+\sqrt{-19})/2]$ be the quadratic integer ring considered at the end of the previous section. We show that the positive field norm $N(a + b(1 + \sqrt{-19})/2) = a^2 + ab + 5b^2$ defined on R is a Dedekind–Hasse norm, which by Proposition 9 and the results of the previous section will prove that R is a P.I.D. but not a Euclidean Domain.

Suppose α, β are nonzero elements of R and $\alpha/\beta \notin R$. We must show that there are elements $s, t \in R$ with $0 < N(s\alpha - t\beta) < N(\beta)$, which by the multiplicativity of the field norm is equivalent to

$$0 < N(\frac{\alpha}{\beta}s - t) < 1. \tag{$*$}$$

Write $\dfrac{\alpha}{\beta} = \dfrac{a + b\sqrt{-19}}{c} \in \mathbb{Q}[\sqrt{-19}]$ with integers a, b, c having no common divisor and with $c > 1$ (since β is assumed not to divide α). Since a, b, c have no common divisor there are integers x, y, z with $ax + by + cz = 1$. Write $ay - 19bx = cq + r$ for some quotient q and remainder r with $|r| \leq c/2$ and let $s = y + x\sqrt{-19}$ and $t = q - z\sqrt{-19}$. Then a quick computation shows that

$$0 < N(\frac{\alpha}{\beta}s - t) = \frac{(ay - 19bx - cq)^2 + 19(ax + by + cz)^2}{c^2} \leq \frac{1}{4} + \frac{19}{c^2}$$

and so $(*)$ is satisfied with this s and t provided $c \geq 5$.

Suppose that $c = 2$. Then one of a, b is even and the other is odd (otherwise $\alpha/\beta \in R$), and then a quick check shows that $s = 1$ and $t = \dfrac{(a-1) + b\sqrt{-19}}{2}$ are elements of R satisfying $(*)$.

Suppose that $c = 3$. The integer $a^2 + 19b^2$ is not divisible by 3 (modulo 3 this is $a^2 + b^2$ which is easily seen to be 0 modulo 3 if and only if a and b are both 0 modulo 3; but then a, b, c have a common factor). Write $a^2 + 19b^2 = 3q + r$ with $r = 1$ or 2. Then again a quick check shows that $s = a - b\sqrt{-19}$, $t = q$ are elements of R satisfying $(*)$.

Finally, suppose that $c = 4$, so a and b are not both even. If one of a, b is even and the other odd, then $a^2 + 19b^2$ is odd, so we can write $a^2 + 19b^2 = 4q + r$ for some $q, r \in \mathbb{Z}$ and $0 < r < 4$. Then $s = a - b\sqrt{-19}$ and $t = q$ satisfy $(*)$. If a and b are both odd, then $a^2 + 19b^2 \equiv 1 + 3 \bmod 8$, so we can write $a^2 + 19b^2 = 8q + 4$ for some $q \in \mathbb{Z}$. Then $s = \dfrac{a - b\sqrt{-19}}{2}$ and $t = q$ are elements of R that satisfy $(*)$.

EXERCISES

1. Prove that in a Principal Ideal Domain two ideals (a) and (b) are comaximal (cf. Section 7.6) if and only if a greatest common divisor of a and b is 1 (in which case a and b are said to be *coprime* or *relatively prime*).
2. Prove that any two nonzero elements of a P.I.D. have a least common multiple (cf. Exercise 11, Section 1).
3. Prove that a quotient of a P.I.D. by a prime ideal is again a P.I.D.
4. Let R be an integral domain. Prove that if the following two conditions hold then R is a Principal Ideal Domain:
   (i) any two nonzero elements a and b in R have a greatest common divisor which can be written in the form $ra + sb$ for some $r, s \in R$, and

(ii) if $a_1, a_2, a_3, \ldots$ are nonzero elements of R such that $a_{i+1} \mid a_i$ for all i, then there is a positive integer N such that a_n is a unit times a_N for all $n \geq N$.

5. Let R be the quadratic integer ring $\mathbb{Z}[\sqrt{-5}]$. Define the ideals $I_2 = (2, 1+\sqrt{-5})$, $I_3 = (3, 2+\sqrt{-5})$, and $I_3' = (3, 2-\sqrt{-5})$.
 (a) Prove that I_2, I_3, and I_3' are nonprincipal ideals in R. [Note that Example 2 following Proposition 1 proves this for I_3.]
 (b) Prove that the product of two nonprincipal ideals can be principal by showing that I_2^2 is the principal ideal generated by 2, i.e., $I_2^2 = (2)$.
 (c) Prove similarly that $I_2 I_3 = (1-\sqrt{-5})$ and $I_2 I_3' = (1+\sqrt{-5})$ are principal. Conclude that the principal ideal (6) is the product of 4 ideals: $(6) = I_2^2 I_3 I_3'$.

6. Let R be an integral domain and suppose that every *prime* ideal in R is principal. This exercise proves that every ideal of R is principal, i.e., R is a P.I.D.
 (a) Assume that the set of ideals of R that are not principal is nonempty and prove that this set has a maximal element under inclusion (which, by hypothesis, is not prime). [Use Zorn's Lemma.]
 (b) Let I be an ideal which is maximal with respect to being nonprincipal, and let $a, b \in R$ with $ab \in I$ but $a \notin I$ and $b \notin I$. Let $I_a = (I, a)$ be the ideal generated by I and a, let $I_b = (I, b)$ be the ideal generated by I and b, and define $J = \{r \in R \mid rI_a \subseteq I\}$. Prove that $I_a = (\alpha)$ and $J = (\beta)$ are principal ideals in R with $I \subsetneq I_b \subseteq J$ and $I_a J = (\alpha\beta) \subseteq I$.
 (c) If $x \in I$ show that $x = s\alpha$ for some $s \in J$. Deduce that $I = I_a J$ is principal, a contradiction, and conclude that R is a P.I.D.

7. An integral domain R in which every ideal generated by two elements is principal (i.e., for every $a, b \in R$, $(a, b) = (d)$ for some $d \in R$) is called a *Bezout Domain*. [cf. also Exercise 11 in Section 3.]
 (a) Prove that the integral domain R is a Bezout Domain if and only if every pair of elements a, b of R has a g.c.d. d in R that can be written as an R-linear combination of a and b, i.e., $d = ax + by$ for some $x, y \in R$.
 (b) Prove that every finitely generated ideal of a Bezout Domain is principal. [cf. the exercises in Sections 9.2 and 9.3 for Bezout Domains in which not every ideal is principal.]
 (c) Let F be the fraction field of the Bezout Domain R. Prove that every element of F can be written in the form a/b with $a, b \in R$ and a and b relatively prime (cf. Exercise 1).

8. Prove that if R is a Principal Ideal Domain and D is a multiplicatively closed subset of R, then $D^{-1}R$ is also a P.I.D. (cf. Section 7.5).

8.3 UNIQUE FACTORIZATION DOMAINS (U.F.D.s)

In the case of the integers $\mathbb{Z}$, there is another method for determining the greatest common divisor of two elements a and b familiar from elementary arithmetic, namely the notion of "factorization into primes" for a and b, from which the greatest common divisor can easily be determined. This can also be extended to a larger class of rings called Unique Factorization Domains (U.F.D.s) — these will be defined shortly. We shall then prove that

every Principal Ideal Domain is a Unique Factorization Domain so that every result about Unique Factorization Domains will automatically hold for both Euclidean Domains and Principal Ideal Domains.

We first introduce some terminology.

Definition. Let R be an integral domain.

(1) Suppose $r \in R$ is nonzero and is not a unit. Then r is called *irreducible* in R if whenever $r = ab$ with $a, b \in R$, at least one of a or b must be a unit in R. Otherwise r is said to be *reducible*.

(2) The nonzero element $p \in R$ is called *prime* in R if the ideal (p) generated by p is a prime ideal. In other words, a nonzero element p is a prime if it is not a unit and whenever $p \mid ab$ for any $a, b \in R$, then either $p \mid a$ or $p \mid b$.

(3) Two elements a and b of R differing by a unit are said to be *associate* in R (i.e., $a = ub$ for some unit u in R).

Proposition 10. In an integral domain a prime element is always irreducible.

Proof: Suppose (p) is a nonzero prime ideal and $p = ab$. Then $ab = p \in (p)$, so by definition of prime ideal one of a or b, say a, is in (p). Thus $a = pr$ for some r. This implies $p = ab = prb$ so $rb = 1$ and b is a unit. This shows that p is irreducible.

It is not true in general that an irreducible element is necessarily prime. For example, consider the element 3 in the quadratic integer ring $R = \mathbb{Z}[\sqrt{-5}\,]$. The computations in Section 1 show that 3 is irreducible in R, but 3 is not a prime since $(2+\sqrt{-5}\,)(2-\sqrt{-5}\,) = 3^2$ is divisible by 3, but neither $2+\sqrt{-5}$ nor $2-\sqrt{-5}$ is divisible by 3 in R.

If R is a Principal Ideal Domain however, the notions of prime and irreducible elements are the same. In particular these notions coincide in $\mathbb{Z}$ and in $F[x]$ (where F is a field).

Proposition 11. In a Principal Ideal Domain a nonzero element is a prime if and only if it is irreducible.

Proof: We have shown above that prime implies irreducible. We must show conversely that if p is irreducible, then p is a prime, i.e., the ideal (p) is a prime ideal. If M is any ideal containing (p) then by hypothesis $M = (m)$ is a principal ideal. Since $p \in (m)$, $p = rm$ for some r. But p is irreducible so by definition either r or m is a unit. This means either $(p) = (m)$ or $(m) = (1)$, respectively. Thus the only ideals containing (p) are (p) or (1), i.e., (p) is a maximal ideal. Since maximal ideals are prime ideals, the proof is complete.

Example

Proposition 11 gives another proof that the quadratic integer ring $\mathbb{Z}[\sqrt{-5}\,]$ is not a P.I.D. since 3 is irreducible but not prime in this ring.

The irreducible elements in the integers $\mathbb{Z}$ are the prime numbers (and their nega-
ves) familiar from elementary arithmetic, and two integers a and b are associates of
e another if and only if $a = \pm b$.

In the integers $\mathbb{Z}$ any integer n can be written as a product of primes (not necessarily stinct), as follows. If n is not itself a prime then by definition it is possible to write $= n_1 n_2$ for two other integers n_1 and n_2 neither of which is a unit, i.e., neither of hich is ± 1. Both n_1 and n_2 must be smaller in absolute value than n itself. If they are th primes, we have already written n as a product of primes. If one of n_1 or n_2 is not ime, then it in turn can be factored into two (smaller) integers. Since integers cannot crease in absolute value indefinitely, we must at some point be left only with prime teger factors, and so we have written n as a product of primes.

For example, if $n = 2210$, the algorithm above proceeds as follows: n is not elf prime, since we can write $n = 2 \cdot 1105$. The integer 2 is a prime, but 1105 is not: $05 = 5 \cdot 221$. The integer 5 is prime, but 221 is not: $221 = 13 \cdot 17$. Here the algorithm rminates, since both 13 and 17 are primes. This gives the *prime factorization* of 2210 $2210 = 2 \cdot 5 \cdot 13 \cdot 17$. Similarly, we find $1131 = 3 \cdot 13 \cdot 29$. In these examples each ime occurs only to the first power, but of course this need not be the case generally.

In the ring $\mathbb{Z}$ not only is it true that every integer n can be written as a product of imes, but in fact this decomposition is *unique* in the sense that any two prime fac-rizations of the same positive integer n differ only in the order in which the positive ime factors are written. The restriction to positive integers is to avoid considering e factorizations $(3)(5)$ and $(-3)(-5)$ of 15 as essentially distinct. This *unique fac-rization* property of $\mathbb{Z}$ (which we shall prove very shortly) is extremely useful for the ithmetic of the integers. General rings with the analogous property are given a name.

efinition. A *Unique Factorization Domain* (U.F.D.) is an integral domain R in which ery nonzero element $r \in R$ which is not a unit has the following two properties:

(i) r can be written as a finite product of irreducibles p_i of R (not necessarily distinct): $r = p_1 p_2 \cdots p_n$ and

(ii) the decomposition in (i) is *unique up to associates*: namely, if $r = q_1 q_2 \cdots q_m$ is another factorization of r into irreducibles, then $m = n$ and there is some renumbering of the factors so that p_i is associate to q_i for $i = 1, 2, \ldots, n$.

xamples

(1) A field F is trivially a Unique Factorization Domain since every nonzero element is a unit, so there are no elements for which properties (i) and (ii) must be verified.

(2) As indicated above, we shall prove shortly that every Principal Ideal Domain is a Unique Factorization Domain (so, in particular, $\mathbb{Z}$ and $F[x]$ where F is a field are both Unique Factorization Domains).

(3) We shall also prove in the next chapter that the ring $R[x]$ of polynomials is a Unique Factorization Domain whenever R itself is a Unique Factorization Domain (in contrast to the properties of being a Principal Ideal Domain or being a Euclidean Domain, which do not carry over from a ring R to the polynomial ring $R[x]$). This result together with the preceding example will show that $\mathbb{Z}[x]$ is a Unique Factorization Domain.

(4) The subring of the Gaussian integers $R = \mathbb{Z}[2i] = \{a + 2bi \mid a, b \in \mathbb{Z}\}$, where $i^2 = -1$, is an integral domain but not a Unique Factorization Domain (rings of this nature were introduced in Exercise 23 of Section 7.1). The elements 2 and $2i$ are irreducibles which are not associates in R since $i \notin R$, and $4 = 2 \cdot 2 = (-2i) \cdot (2i)$ has two distinct factorizations in R. One may also check directly that $2i$ is irreducible but not prime in R (since $R/(2i) \cong \mathbb{Z}/4\mathbb{Z}$). In the larger ring of Gaussian integers, $\mathbb{Z}[i]$, (which is a Unique Factorization Domain) 2 and $2i$ *are* associates since i is a unit in this larger ring. We shall give a slightly different proof that $\mathbb{Z}[2i]$ is not a Unique Factorization Domain at the end of Section 9.3 (one in which we do not have to check that 2 and $2i$ are irreducibles).

(5) The quadratic integer ring $\mathbb{Z}[\sqrt{-5}\,]$ is another example of an integral domain that is not a Unique Factorization Domain, since $6 = 2 \cdot 3 = (1 + \sqrt{-5}\,)(1 - \sqrt{-5}\,)$ gives two distinct factorizations of 6 into irreducibles. The principal ideal (6) in $\mathbb{Z}[\sqrt{-5}\,]$ can be written as a product of 4 nonprincipal prime ideals: $(6) = P_2^2 P_3 P_3'$ and the two distinct factorizations of the element 6 in $\mathbb{Z}[\sqrt{-5}\,]$ can be interpreted as arising from two rearrangements of this product of ideals into products of principal ideals: the product of $P_2^2 = (2)$ with $P_3 P_3' = (3)$, and the product of $P_2 P_3 = (1 + \sqrt{-5}\,)$ with $P_2 P_3' = (1 - \sqrt{-5}\,)$ (cf. Exercise 8).

While the *elements* of the quadratic integer ring $\mathcal{O}$ need not have unique factorization, it is a theorem (Corollary 16.16) that every *ideal* in $\mathcal{O}$ can be written uniquely as a product of prime *ideals*. The unique factorization of ideals into the product of prime ideals holds in general for rings of integers of algebraic number fields (examples of which are the quadratic integer rings) and leads to the notion of a Dedekind Domain considered in Chapter 16. It was the failure to have unique factorization into irreducibles for elements in algebraic integer rings in number theory that originally led to the definition of an ideal. The resulting uniqueness of the decomposition into prime ideals in these rings gave the elements of the ideals an "ideal" (in the sense of "perfect" or "desirable") behavior that is the basis for the choice of terminology for these (now fundamental) algebraic objects.

The first property of irreducible elements in a Unique Factorization Domain is that they are also primes. One might think that we could deduce Proposition 11 from this proposition together with the previously mentioned theorem (that we shall prove shortly) that every Principal Ideal Domain is a Unique Factorization Domain, however Proposition 11 will be used in the proof of the latter theorem.

Proposition 12. In a Unique Factorization Domain a nonzero element is a prime if and only if it is irreducible.

Proof: Let R be a Unique Factorization Domain. Since by Proposition 10, primes of R are irreducible it remains to prove that each irreducible element is a prime. Let p be an irreducible in R and assume $p \mid ab$ for some $a, b \in R$; we must show that p divides either a or b. To say that p divides ab is to say $ab = pc$ for some c in R. Writing a and b as a product of irreducibles, we see from this last equation and from the *uniqueness* of the decomposition into irreducibles of ab that the irreducible element p must be *associate* to one of the irreducibles occurring either in the factorization of a or in the factorization of b. We may assume that p is associate to one of the irreducibles in the factorization of a, i.e., that a can be written as a product $a = (up)p_2 \cdots p_n$ for u a unit and some (possibly empty set of) irreducibles $p_2, \ldots, p_n$. But then p divides a, since $a = pd$ with $d = up_2 \cdots p_n$, completing the proof.

In a Unique Factorization Domain we shall now use the terms "prime" and "irreducible" interchangeably although we shall usually refer to the "primes" in $\mathbb{Z}$ and the "irreducibles" in $F[x]$.

We shall use the preceding proposition to show that in a Unique Factorization Domain any two nonzero elements a and b have a greatest common divisor:

Proposition 13. Let a and b be two nonzero elements of the Unique Factorization Domain R and suppose

$$a = up_1{}^{e_1}p_2{}^{e_2}\cdots p_n{}^{e_n} \qquad \text{and} \qquad b = vp_1{}^{f_1}p_2{}^{f_2}\cdots p_n{}^{f_n}$$

are prime factorizations for a and b, where u and v are units, the primes $p_1, p_2, \dots, p_n$ are *distinct* and the exponents e_i and f_i are ≥ 0. Then the element

$$d = p_1{}^{\min(e_1, f_1)}p_2{}^{\min(e_2, f_2)}\cdots p_n{}^{\min(e_n, f_n)}$$

(where $d = 1$ if all the exponents are 0) is a greatest common divisor of a and b.

Proof: Since the exponents of each of the primes occurring in d are no larger than the exponents occurring in the factorizations of both a and b, d divides both a and b. To show that d is a greatest common divisor, let c be any common divisor of a and b and let $c = q_1{}^{g_1}q_2{}^{g_2}\cdots q_m{}^{g_m}$ be the prime factorization of c. Since each q_i divides c, hence divides a and b, we see from the preceding proposition that q_i must divide one of the primes p_j. In particular, up to associates (so up to multiplication by a unit) the primes occurring in c must be a subset of the primes occurring in a and b : $\{q_1, q_2, \dots, q_m\} \subseteq \{p_1, p_2, \dots, p_n\}$. Similarly, the exponents for the primes occurring in c must be no larger than those occurring in d. This implies that c divides d, completing the proof.

Example

In the example above, where $a = 2210$ and $b = 1131$, we find immediately from their prime factorizations that $(a, b) = 13$. Note that if the prime factorizations for a and b are known, the proposition above gives their greatest common divisor instantly, but that finding these prime factorizations is extremely time-consuming computationally. The Euclidean Algorithm is the fastest method for determining the g.c.d. of two integers but unfortunately it gives almost no information on the prime factorizations of the integers.

We now come to one of the principal results relating some of the rings introduced in this chapter.

Theorem 14. Every Principal Ideal Domain is a Unique Factorization Domain. In particular, every Euclidean Domain is a Unique Factorization Domain.

Proof: Note that the second assertion follows from the first since Euclidean Domains are Principal Ideal Domains. To prove the first assertion let R be a Principal Ideal Domain and let r be a nonzero element of R which is not a unit. We must show first that r can be written as a finite product of irreducible elements of R and then we must verify that this decomposition is unique up to units.

The method of proof of the first part is precisely analogous to the determination of the prime factor decomposition of an integer. Assume r is nonzero and is not a unit. If r is itself irreducible, then we are done. If not, then by definition r can be written as a product $r = r_1 r_2$ where neither r_1 nor r_2 is a unit. If both these elements are irreducibles, then again we are done, having written r as a product of irreducible elements. Otherwise, at least one of the two elements, say r_1 is reducible, hence can be written as a product of two nonunit elements $r_1 = r_{11} r_{12}$, and so forth. What we must verify is that this process *terminates*, i.e., that we must necessarily reach a point where all of the elements obtained as factors of r are irreducible. Suppose this is not the case. From the factorization $r = r_1 r_2$ we obtain a *proper* inclusion of ideals: $(r) \subset (r_1) \subset R$. The first inclusion is proper since r_2 is not a unit, and the last inclusion is proper since r_1 is not a unit. From the factorization of r_1 we similarly obtain $(r) \subset (r_1) \subset (r_{11}) \subset R$. If this process of factorization did not terminate after a finite number of steps, then we would obtain an *infinite ascending chain* of ideals:

$$(r) \subset (r_1) \subset (r_{11}) \subset \cdots \subset R$$

where all containments are proper, and the Axiom of Choice ensures that an infinite chain exists (cf. Appendix I).

We now show that any ascending chain $I_1 \subseteq I_2 \subseteq \cdots \subseteq R$ of ideals in a Principal Ideal Domain eventually becomes stationary, i.e., there is some positive integer n such that $I_k = I_n$ for all $k \geq n$.[3] In particular, it is not possible to have an infinite ascending chain of ideals where all containments are proper. Let $I = \cup_{i=1}^{\infty} I_i$. It follows easily (as in the proof of Proposition 11 in Section 7.4) that I is an ideal. Since R is a Principal Ideal Domain it is principally generated, say $I = (a)$. Since I is the union of the ideals above, a must be an element of one of the ideals in the chain, say $a \in I_n$. But then we have $I_n \subseteq I = (a) \subseteq I_n$ and so $I = I_n$ and the chain becomes stationary at I_n. This proves that every nonzero element of R which is not a unit has some factorization into irreducibles in R.

It remains to prove that the above decomposition is essentially unique. We proceed by induction on the number, n, of irreducible factors in some factorization of the element r. If $n = 0$, then r is a unit. If we had $r = qc$ (some other factorization) for some irreducible q, then q would divide a unit, hence would itself be a unit, a contradiction. Suppose now that n is at least 1 and that we have two products

$$r = p_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m \qquad m \geq n$$

for r where the p_i and q_j are (not necessarily distinct) irreducibles. Since then p_1 divides the product on the right, we see by Proposition 11 that p_1 must divide one of the factors. Renumbering if necessary, we may assume p_1 divides q_1. But then $q_1 = p_1 u$ for some element u of R which must in fact be a unit since q_1 is irreducible. Thus p_1 and q_1 are associates. Cancelling p_1 (recall we are in an integral domain, so this is legitimate), we obtain the equation

$$p_2 \cdots p_n = u q_2 q_3 \cdots q_m = q_2' q_3 \cdots q_m \qquad m \geq n.$$

[3]This same argument can be used to prove the more general statement: an ascending chain of ideals becomes stationary in any commutative ring where all the ideals are *finitely generated*. This result will be needed in Chapter 12 where the details will be repeated.

where $q_2' = uq_2$ is again an irreducible (associate to q_2). By induction on n, we conclude that each of the factors on the left matches bijectively (up to associates) with the factors on the far right, hence with the factors in the middle (which are the same, up to associates). Since p_1 and q_1 (after the initial renumbering) have already been shown to be associate, this completes the induction step and the proof of the theorem.

Corollary 15. *(Fundamental Theorem of Arithmetic)* The integers $\mathbb{Z}$ are a Unique Factorization Domain.

Proof: The integers $\mathbb{Z}$ are a Euclidean Domain, hence are a Unique Factorization Domain by the theorem.

We can now complete the equivalence (Proposition 9) between the existence of a Dedekind–Hasse norm on the integral domain R and whether R is a P.I.D.

Corollary 16. Let R be a P.I.D. Then there exists a multiplicative Dedekind–Hasse norm on R.

Proof: If R is a P.I.D. then R is a U.F.D. Define the norm N by setting $N(0) = 0$, $N(u) = 1$ if u is a unit, and $N(a) = 2^n$ if $a = p_1 p_2 \cdots p_n$ where the p_i are irreducibles in R (well defined since the number of irreducible factors of a is unique). Clearly $N(ab) = N(a)N(b)$ so N is positive and multiplicative. To show that N is a Dedekind–Hasse norm, suppose that a, b are nonzero elements of R. Then the ideal generated by a and b is principal by assumption, say $(a, b) = (r)$. If a is not contained in the ideal (b) then also r is not contained in (b), i.e., r is not divisible by b. Since $b = xr$ for some $x \in R$, it follows that x is not a unit in R and so $N(b) = N(x)N(r) > N(r)$. Hence (a, b) contains a nonzero element with norm strictly smaller than the norm of b, completing the proof.

Factorization in the Gaussian Integers

We end our discussion of Unique Factorization Domains by describing the irreducible elements in the Gaussian integers $\mathbb{Z}[i]$ and the corresponding application to a famous theorem of Fermat in elementary number theory. This is particularly appropriate since the classical study of $\mathbb{Z}[i]$ initiated the algebraic study of rings.

In general, let $\mathcal{O}$ be a quadratic integer ring and let N be the associated field norm introduced in Section 7.1. Suppose $\alpha \in \mathcal{O}$ is an element whose norm is a prime p in $\mathbb{Z}$. If $\alpha = \beta\gamma$ for some $\beta, \gamma \in \mathcal{O}$ then $p = N(\alpha) = N(\beta)N(\gamma)$ so that one of $N(\beta)$ or $N(\gamma)$ is ± 1 and the other is $\pm p$. Since we have seen that an element of $\mathcal{O}$ has norm ± 1 if and only if it is a unit in $\mathcal{O}$, one of the factors of α is a unit. It follows that

if $N(\alpha)$ is $\pm$ a prime (in $\mathbb{Z}$), then α is irreducible in $\mathcal{O}$.

Suppose that π is a prime element in $\mathcal{O}$ and let (π) be the ideal generated by π in $\mathcal{O}$. Since (π) is a prime ideal in $\mathcal{O}$ it is easy to check that $(\pi) \cap \mathbb{Z}$ is a prime ideal in $\mathbb{Z}$ (if a and b are integers with $ab \in (\pi)$ then either a or b is an element of (π), so a or b is in $(\pi) \cap \mathbb{Z}$). Since $N(\pi)$ is a nonzero integer in (π) we have $(\pi) \cap \mathbb{Z} = p\mathbb{Z}$ for some integer prime p. It follows from $p \in (\pi)$ that π is a divisor in $\mathcal{O}$ of the integer prime p, and so the prime elements in $\mathcal{O}$ can be found by determining how the primes in $\mathbb{Z}$ factor in the larger ring $\mathcal{O}$. Suppose π divides the prime p in $\mathcal{O}$, say $p = \pi\pi'$. Then $N(\pi)N(\pi') = N(p) = p^2$, so since π is not a unit there are only two possibilities: either $N(\pi) = \pm p^2$ or $N(\pi) = \pm p$. In the former case $N(\pi') = \pm 1$, hence π' is a unit and $p = \pi$ (up to associates) is irreducible in $\mathbb{Z}[i]$. In the latter case $N(\pi) = N(\pi') = \pm p$, hence π' is also irreducible and $p = \pi\pi'$ is the product of precisely two irreducibles.

Consider now the special case $D = -1$ of the Gaussian integers $\mathbb{Z}[i]$. We have seen that the units in $\mathbb{Z}[i]$ are the elements ± 1 and $\pm i$. We proved in Section 1 that $\mathbb{Z}[i]$ is a Euclidean Domain, hence is also a Principal Ideal Domain and a Unique Factorization Domain, so the irreducible elements are the same as the prime elements, and can be determined by seeing how the primes in $\mathbb{Z}$ factor in the larger ring $\mathbb{Z}[i]$.

In this case $\alpha = a+bi$ has $N(\alpha) = \alpha\overline{\alpha} = a^2+b^2$, where $\overline{\alpha} = a-bi$ is the complex conjugate of α. It follows by what we just saw that *p factors in $\mathbb{Z}[i]$ into precisely two irreducibles if and only if $p = a^2 + b^2$ is the sum of two integer squares (otherwise p remains irreducible in $\mathbb{Z}[i]$).* If $p = a^2 + b^2$ then the corresponding irreducible elements in $\mathbb{Z}[i]$ are $a \pm bi$.

Clearly $2 = 1^2 + 1^2$ is the sum of two squares, giving the factorization $2 = (1+i)(1-i) = -i(1+i)^2$. The irreducibles $1+i$ and $1-i = -i(1+i)$ are associates and it is easy to check that this is the only situation in which conjugate irreducibles $a + bi$ and $a - bi$ can be associates.

Since the square of any integer is congruent to either 0 or 1 modulo 4, an odd prime in $\mathbb{Z}$ that is the sum of two squares must be congruent to 1 modulo 4. Thus if p is a prime of $\mathbb{Z}$ with $p \equiv 3 \bmod 4$ then p is not the sum of two squares and p remains irreducible in $\mathbb{Z}[i]$.

Suppose now that p is a prime of $\mathbb{Z}$ with $p \equiv 1 \bmod 4$. We shall prove that p cannot be irreducible in $\mathbb{Z}[i]$ which will show that $p = (a+bi)(a-bi)$ factors as the product of two distinct irreducibles in $\mathbb{Z}[i]$ or, equivalently, that $p = a^2 + b^2$ is the sum of two squares. We first prove the following result from elementary number theory:

Lemma 17. The prime number $p \in \mathbb{Z}$ divides an integer of the form $n^2 + 1$ if and only if p is either 2 or is an odd prime congruent to 1 modulo 4.

Proof: The statement for $p = 2$ is trivial since $2 \mid 1^2 + 1$. If p is an odd prime, note that $p \mid n^2 + 1$ is equivalent to $n^2 = -1$ in $\mathbb{Z}/p\mathbb{Z}$. This in turn is equivalent to saying the residue class of n is of order 4 in the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$. Thus p divides an integer of the form $n^2 + 1$ if and only if $(\mathbb{Z}/p\mathbb{Z})^\times$ contains an element of order 4. By Lagrange's Theorem, if $(\mathbb{Z}/p\mathbb{Z})^\times$ contains an element of order 4 then $|(\mathbb{Z}/p\mathbb{Z})^\times| = p - 1$ is divisible by 4, i.e., p is congruent to 1 modulo 4.

Conversely, suppose $p - 1$ is divisible by 4. We first argue that $(\mathbb{Z}/p\mathbb{Z})^\times$ contains a unique element of order 2. If $m^2 \equiv 1 \bmod p$ then p divides $m^2 - 1 = (m-1)(m+1)$. Thus p divides either $m - 1$ (i.e., $m \equiv 1 \bmod p$) or $m + 1$ (i.e., $m \equiv -1 \bmod p$), so -1 is the unique residue class of order 2 in $(\mathbb{Z}/p\mathbb{Z})^\times$. Now the abelian group $(\mathbb{Z}/p\mathbb{Z})^\times$ contains a subgroup H of order 4 (for example, the quotient by the subgroup $\{\pm 1\}$ contains a subgroup of order 2 whose preimage is a subgroup of order 4 in $(\mathbb{Z}/p\mathbb{Z})^\times$).

Since the Klein 4-group has three elements of order 2 whereas $(\mathbb{Z}/p\mathbb{Z})^\times$ — hence also H — has a unique element of order 2, H must be the cyclic group of order 4. Thus $(\mathbb{Z}/p\mathbb{Z})^\times$ contains an element of order 4, namely a generator for H.

Remark: We shall prove later (Corollary 19 in Section 9.5) that $(\mathbb{Z}/p\mathbb{Z})^\times$ is a cyclic group, from which it is immediate that there is an element of order 4 if and only if $p-1$ is divisible by 4.

By Lemma 17, if $p \equiv 1 \bmod 4$ is a prime then p divides n^2+1 in $\mathbb{Z}$ for some $n \in \mathbb{Z}$, so certainly p divides $n^2+1 = (n+i)(n-i)$ in $\mathbb{Z}[i]$. If p were irreducible in $\mathbb{Z}[i]$ then p would divide either $n+i$ or $n-i$ in $\mathbb{Z}[i]$. In this situation, since p is a real number, it would follow that p divides both $n+i$ and its complex conjugate $n-i$; hence p would divide their difference, $2i$. This is clearly not the case. We have proved the following result:

Proposition 18.

(1) *(Fermat's Theorem on sums of squares)* The prime p is the sum of two integer squares, $p = a^2+b^2$, $a, b \in \mathbb{Z}$, if and only if $p = 2$ or $p \equiv 1 \bmod 4$. Except for interchanging a and b or changing the signs of a and b, the representation of p as a sum of two squares is unique.

(2) The irreducible elements in the Gaussian integers $\mathbb{Z}[i]$ are as follows:
   - (a) $1+i$ (which has norm 2),
   - (b) the primes $p \in \mathbb{Z}$ with $p \equiv 3 \bmod 4$ (which have norm p^2), and
   - (c) $a+bi$, $a-bi$, the distinct irreducible factors of $p = a^2+b^2 = (a+bi)(a-bi)$ for the primes $p \in \mathbb{Z}$ with $p \equiv 1 \bmod 4$ (both of which have norm p).

The first part of Proposition 18 is a famous theorem of Fermat in elementary number theory, for which a number of alternate proofs can be given.

More generally, the question of whether the integer $n \in \mathbb{Z}$ can be written as a sum of two integer squares, $n = A^2+B^2$, is equivalent to the question of whether n is the norm of an element $A+Bi$ in the Gaussian integers, i.e., $n = A^2+B^2 = N(A+Bi)$. Writing $A+Bi = \pi_1\pi_2\cdots\pi_k$ as a product of irreducibles (uniquely up to units) it follows from the explicit description of the irreducibles in $\mathbb{Z}[i]$ in Proposition 18 that n is a norm if and only if the prime divisors of n that are congruent to 3 mod 4 occur to even exponents. Further, if this condition on n is satisfied, then the uniqueness of the factorization of $A+Bi$ in $\mathbb{Z}[i]$ allows us to count the number of representations of n as a sum of two squares, as in the following corollary.

Corollary 19. Let n be a positive integer and write

$$n = 2^k p_1^{a_1} \cdots p_r^{a_r} q_1^{b_1} \cdots q_s^{b_s}$$

where $p_1, \ldots, p_r$ are distinct primes congruent to 1 modulo 4 and $q_1, \ldots, q_s$ are distinct primes congruent to 3 modulo 4. Then n can be written as a sum of two squares in $\mathbb{Z}$, i.e., $n = A^2+B^2$ with $A, B \in \mathbb{Z}$, if and only if each b_i is even. Further, if this condition on n is satisfied, then the number of representations of n as a sum of two squares is $4(a_1+1)(a_2+1)\cdots(a_r+1)$.

Proof: The first statement in the corollary was proved above. Assume now that $b_1, \ldots, b_s$ are all even. For each prime p_i congruent to 1 modulo 4 write $p_i = \pi_i \overline{\pi_i}$ for $i = 1, 2, \ldots, r$, where π_i and $\overline{\pi_i}$ are irreducibles as in (2)(c) of Proposition 18. If $N(A + Bi) = n$ then examining norms we see that, up to units, the factorization of $A + Bi$ into irreducibles in $\mathbb{Z}[i]$ is given by

$$A + Bi = (1+i)^k (\pi_1^{a_{1,1}} \overline{\pi_1}^{a_{1,2}}) \ldots (\pi_r^{a_{r,1}} \overline{\pi_r}^{a_{r,2}}) q_1^{b_1/2} \ldots q_s^{b_s/2}$$

with nonnegative integers $a_{i,1}, a_{i,2}$ satisfying $a_{i,1} + a_{i,2} = a_i$ for $i = 1, 2, \ldots, r$. Since $a_{i,1}$ can have the values $0, 1, \ldots, a_i$ (and then $a_{i,2}$ is determined), there are a total of $(a_1 + 1)(a_2 + 1) \cdots (a_r + 1)$ distinct elements $A + Bi$ in $\mathbb{Z}[i]$ of norm n, up to units. Finally, since there are four units in $\mathbb{Z}[i]$, the second statement in the corollary follows.

Example

Since $493 = 17 \cdot 29$ and both primes are congruent to 1 modulo 4, $493 = A^2 + B^2$ is the sum of two integer squares. Since $17 = (4+i)(4-i)$ and $29 = (5+2i)(5-2i)$ the possible factorizations of $A + Bi$ in $\mathbb{Z}[i]$ up to units are $(4+i)(5+2i) = 18 + 13i$, $(4+i)(5-2i) = 22 - 3i$, $(4-i)(5-2i) = 22 + 3i$, and $(4-i)(5-2i) = 18 - 13i$. Multiplying by -1 reverses both signs and multiplication by i interchanges the A and B and introduces one sign change. Then $493 = (\pm 18)^2 + (\pm 13)^2 = (\pm 22)^2 + (\pm 3)^2$ with all possible choices of signs give 8 of the 16 possible representations of 493 as the sum of two squares; the remaining 8 are obtained by interchanging the two summands.

Similarly, the integer $58000957 = 7^6 \cdot 17 \cdot 29$ can be written as a sum of two squares in precisely 16 ways, obtained by multiplying each of the integers A, B in $493 = A^2 + B^2$ above by 7^3.

Summary

In summary, we have the following inclusions among classes of commutative rings with identity:

fields $\subset$ *Euclidean Domains* $\subset$ *P.I.D.s* $\subset$ *U.F.D.s* $\subset$ *integral domains*

with all containments being proper. Recall that $\mathbb{Z}$ is a Euclidean Domain that is not a field, the quadratic integer ring $\mathbb{Z}[(1 + \sqrt{-19})/2]$ is a Principal Ideal Domain that is not a Euclidean Domain, $\mathbb{Z}[x]$ is a Unique Factorization Domain (Theorem 7 in Chapter 9) that is not a Principal Ideal Domain and $\mathbb{Z}[\sqrt{-5}\,]$ is an integral domain that is not a Unique Factorization Domain.

EXERCISES

1. Let $G = \mathbb{Q}^\times$ be the multiplicative group of nonzero rational numbers. If $\alpha = p/q \in G$, where p and q are relatively prime integers, let $\varphi : G \to G$ be the map which interchanges the primes 2 and 3 in the prime power factorizations of p and q (so, for example, $\varphi(2^4 3^{11} 5^1 13^2) = 3^4 2^{11} 5^1 13^2$, $\varphi(3/16) = \varphi(3/2^4) = 2/3^4 = 2/81$, and φ is the identity on all rational numbers with numerators and denominators relatively prime to 2 and to 3).
   (a) Prove that φ is a group isomorphism.
   (b) Prove that there are infinitely many isomorphisms of the group G to itself.

(c) Prove that none of the isomorphisms above can be extended to an isomorphism of the *ring* $\mathbb{Q}$ to itself. In fact prove that the identity map is the only ring isomorphism of $\mathbb{Q}$.

2. Let a and b be nonzero elements of the Unique Factorization Domain R. Prove that a and b have a least common multiple (cf. Exercise 11 of Section 1) and describe it in terms of the prime factorizations of a and b in the same fashion that Proposition 13 describes their greatest common divisor.

3. Determine all the representations of the integer $2130797 = 17^2 \cdot 73 \cdot 101$ as a sum of two squares.

4. Prove that if an integer is the sum of two rational squares, then it is the sum of two integer squares (for example, $13 = (1/5)^2 + (18/5)^2 = 2^2 + 3^2$).

5. Let $R = \mathbb{Z}[\sqrt{-n}]$ where n is a squarefree integer greater than 3.
   (a) Prove that 2, $\sqrt{-n}$ and $1 + \sqrt{-n}$ are irreducibles in R.
   (b) Prove that R is not a U.F.D. Conclude that the quadratic integer ring $\mathcal{O}$ is not a U.F.D. for $D \equiv 2, 3 \bmod 4$, $D < -3$ (so also not Euclidean and not a P.I.D.). [Show that either $\sqrt{-n}$ or $1 + \sqrt{-n}$ is not prime.]
   (c) Give an explicit ideal in R that is not principal. [Using (b) consider a maximal ideal containing the nonprime ideal $(\sqrt{-n})$ or $(1 + \sqrt{-n})$.]

6. (a) Prove that the quotient ring $\mathbb{Z}[i]/(1+i)$ is a field of order 2.
   (b) Let $q \in \mathbb{Z}$ be a prime with $q \equiv 3 \bmod 4$. Prove that the quotient ring $\mathbb{Z}[i]/(q)$ is a field with q^2 elements.
   (c) Let $p \in \mathbb{Z}$ be a prime with $p \equiv 1 \bmod 4$ and write $p = \pi\bar{\pi}$ as in Proposition 18. Show that the hypotheses for the Chinese Remainder Theorem (Theorem 17 in Section 7.6) are satisfied and that $\mathbb{Z}[i]/(p) \cong \mathbb{Z}[i]/(\pi) \times \mathbb{Z}[i]/(\bar{\pi})$ as rings. Show that the quotient ring $\mathbb{Z}[i]/(p)$ has order p^2 and conclude that $\mathbb{Z}[i]/(\pi)$ and $\mathbb{Z}[i]/(\bar{\pi})$ are both fields of order p.

7. Let π be an irreducible element in $\mathbb{Z}[i]$.
   (a) For any integer $n \geq 0$, prove that $(\pi^{n+1}) = \pi^{n+1}\mathbb{Z}[i]$ is an ideal in $(\pi^n) = \pi^n\mathbb{Z}[i]$ and that multiplication by π^n induces an isomorphism $\mathbb{Z}[i]/(\pi) \cong (\pi^n)/(\pi^{n+1})$ as additive abelian groups.
   (b) Prove that $|\mathbb{Z}[i]/(\pi^n)| = |\mathbb{Z}[i]/(\pi)|^n$.
   (c) Prove for any nonzero α in $\mathbb{Z}[i]$ that the quotient ring $\mathbb{Z}[i]/(\alpha)$ has order equal to $N(\alpha)$. [Use (b) together with the Chinese Remainder Theorem and the results of the previous exercise.]

8. Let R be the quadratic integer ring $\mathbb{Z}[\sqrt{-5}]$ and define the ideals $I_2 = (2, 1 + \sqrt{-5})$, $I_3 = (3, 2 + \sqrt{-5})$, and $I_3' = (3, 2 - \sqrt{-5})$.
   (a) Prove that 2, 3, $1 + \sqrt{-5}$ and $1 - \sqrt{-5}$ are irreducibles in R, no two of which are associate in R, and that $6 = 2 \cdot 3 = (1 + \sqrt{-5}) \cdot (1 - \sqrt{-5})$ are two distinct factorizations of 6 into irreducibles in R.
   (b) Prove that I_2, I_3, and I_3' are prime ideals in R. [One approach: for I_3, observe that $R/I_3 \cong (R/(3))/(I_3/(3))$ by the Third Isomorphism Theorem for Rings. Show that $R/(3)$ has 9 elements, $(I_3/(3))$ has 3 elements, and that $R/I_3 \cong \mathbb{Z}/3\mathbb{Z}$ as an additive abelian group. Conclude that I_3 is a maximal (hence prime) ideal and that $R/I_3 \cong \mathbb{Z}/3\mathbb{Z}$ as rings.]
   (c) Show that the factorizations in (a) imply the equality of ideals $(6) = (2)(3)$ and $(6) = (1 + \sqrt{-5})(1 - \sqrt{-5})$. Show that these two ideal factorizations give the same factorization of the ideal (6) as the product of prime ideals (cf. Exercise 5 in Section 2).

9. Suppose that the quadratic integer ring $\mathcal{O}$ is a P.I.D. Prove that the absolute value of the field norm N on $\mathcal{O}$ (cf. Section 7.1) is a Dedekind–Hasse norm on $\mathcal{O}$. Conclude that if the quadratic integer ring $\mathcal{O}$ possesses *any* Dedekind–Hasse norm, then in fact the absolute value of the field norm on $\mathcal{O}$ already provides a Dedekind–Hasse norm on $\mathcal{O}$. [If $\alpha, \beta \in \mathcal{O}$ then $(\alpha, \beta) = (\gamma)$ for some $\gamma \in \mathcal{O}$. Show that if β does not divide α then $0 < |N(\gamma)| < |N(\beta)|$ — use the fact that the units in $\mathcal{O}$ are precisely the elements whose norm is ± 1.]

Remark: If $\mathcal{O}$ is a Euclidean Domain with respect to some norm it is not necessarily true that it is a Euclidean Domain with respect to the absolute value of the field norm (although this is true for $D < 0$, cf. Exercise 8 in Section 1). An example is $D = 69$ (cf. D. Clark, *A quadratic field which is Euclidean but not norm-Euclidean*, Manuscripta Math., 83(1994), pp. 327–330).

10. (*k-stage Euclidean Domains*) Let R be an integral domain and let $N : R \to \mathbb{Z}^+ \cup \{0\}$ be a norm on R. The ring R is Euclidean with respect to N if for any $a, b \in R$ with $b \neq 0$, there exist elements q and r in R with

$$a = qb + r \qquad \text{with } r = 0 \text{ or } N(r) < N(b).$$

Suppose now that this condition is weakened, namely that for any $a, b \in R$ with $b \neq 0$, there exist elements q, q' and r, r' in R with

$$a = qb + r, \quad b = q'r + r' \qquad \text{with } r' = 0 \text{ or } N(r') < N(b),$$

i.e., the remainder after two divisions is smaller. Call such a domain a *2-stage Euclidean Domain*.

(a) Prove that iterating the divisions in a 2-stage Euclidean Domain produces a greatest common divisor of a and b which is a linear combination of a and b. Conclude that every *finitely generated* ideal of a 2-stage Euclidean Domain is principal. (There are 2-stage Euclidean Domains that are *not* P.I.D.s, however.) [Imitate the proof of Theorem 4.]

(b) Prove that a 2-stage Euclidean Domain in which every nonzero nonunit can be factored into a finite number of irreducibles is a Unique Factorization Domain. [Prove first that irreducible elements are prime, as follows. If p is irreducible and $p \mid ab$ with p not dividing a, use part (a) to write $px + ay = 1$ for some x, y. Multiply through by b to conclude that $p \mid b$, so p is prime. Now follow the proof of uniqueness in Theorem 14.]

(c) Make the obvious generalization to define the notion of a k-stage Euclidean Domain for any integer $k \geq 1$. Prove that statements (a) and (b) remain valid if "2-stage Euclidean" is replaced by "k-stage Euclidean."

Remarks: There are examples of rings which are 2-stage Euclidean but are not Euclidean. There are also examples of rings which are not Euclidean with respect to a given norm but which are k-stage Euclidean with respect to the norm (for example, the ring $\mathbb{Z}[\sqrt{14}]$ is not Euclidean with respect to the usual norm $N(a+b\sqrt{14}) = |a^2 - 14b^2|$, but is 2-stage Euclidean with respect to this norm). The k-stage Euclidean condition is also related to the question of whether the group $GL_n(R)$ of invertible $n \times n$ matrices with entries from R is generated by elementary matrices (matrices with 1's along the main diagonal, a single 1 somewhere off the main diagonal, and 0's elsewhere).

11. (*Characterization of P.I.D.s*) Prove that R is a P.I.D. if and only if R is a U.F.D. that is also a Bezout Domain (cf. Exercise 7 in Section 2). [One direction is given by Theorem 14. For the converse, let a be a nonzero element of the ideal I with a minimal number of irreducible factors. Prove that $I = (a)$ by showing that if there is an element $b \in I$ that is not in (a) then $(a, b) = (d)$ leads to a contradiction.]

CHAPTER 9

Polynomial Rings

We begin this chapter on polynomial rings with a summary of facts from the preceding two chapters (with references where needed). The basic definitions were given in slightly greater detail in Section 7.2. For convenience, the ring R will always be a commutative ring with identity $1 \neq 0$.

9.1 DEFINITIONS AND BASIC PROPERTIES

The polynomial ring $R[x]$ in the indeterminate x with coefficients from R is the set of all formal sums $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ with $n \geq 0$ and each $a_i \in R$. If $a_n \neq 0$ then the polynomial is of degree n, $a_n x^n$ is the leading term, and a_n is the leading coefficient (where the leading coefficient of the zero polynomial is defined to be 0). The polynomial is monic if $a_n = 1$. Addition of polynomials is "componentwise":

$$\sum_{i=0}^{n} a_i x^i + \sum_{i=0}^{n} b_i x^i = \sum_{i=0}^{n} (a_i + b_i) x^i$$

(here a_n or b_n may be zero in order for addition of polynomials of different degrees to be defined). Multiplication is performed by first defining $(ax^i)(bx^j) = abx^{i+j}$ and then extending to all polynomials by the distributive laws so that in general

$$\left(\sum_{i=0}^{n} a_i x^i\right) \times \left(\sum_{i=0}^{m} b_i x^i\right) = \sum_{k=0}^{n+m} \left(\sum_{i=0}^{k} a_i b_{k-i}\right) x^k.$$

In this way $R[x]$ is a commutative ring with identity (the identity 1 from R) in which we identify R with the subring of constant polynomials.

We have already noted that if R is an integral domain then the leading term of a product of polynomials is the product of the leading terms of the factors. The following is Proposition 4 of Section 7.2 which we record here for completeness.

Proposition 1. Let R be an integral domain. Then

(1) degree $p(x)q(x) =$ degree $p(x) +$ degree $q(x)$ if $p(x), q(x)$ are nonzero

(2) the units of $R[x]$ are just the units of R

(3) $R[x]$ is an integral domain.

Recall also that if R is an integral domain, the quotient field of $R[x]$ consists of all quotients $\frac{p(x)}{q(x)}$ where $q(x)$ is not the zero polynomial (and is called the field of rational functions in x with coefficients in R).

The next result describes a relation between the ideals of R and those of $R[x]$.

Proposition 2. Let I be an ideal of the ring R and let $(I) = I[x]$ denote the ideal of $R[x]$ generated by I (the set of polynomials with coefficients in I). Then

$$R[x]/(I) \cong (R/I)[x].$$

In particular, if I is a prime ideal of R then (I) is a prime ideal of $R[x]$.

Proof: There is a natural map $\varphi : R[x] \to (R/I)[x]$ given by reducing each of the coefficients of a polynomial modulo I. The definition of addition and multiplication in these two rings shows that φ is a ring homomorphism. The kernel is precisely the set of polynomials each of whose coefficients is an element of I, which is to say that $\ker \varphi = I[x] = (I)$, proving the first part of the proposition. The last statement follows from Proposition 1, since if I is a prime ideal in R, then R/I is an integral domain, hence also $(R/I)[x]$ is an integral domain. This shows if I is a prime ideal of R, then (I) is a prime ideal of $R[x]$.

Note that it is not true that if I is a maximal ideal of R then (I) is a maximal ideal of $R[x]$. However, if I is maximal in R then the ideal of $R[x]$ generated by I and x is maximal in $R[x]$.

We now give an example of the "reduction homomorphism" of Proposition 2 which will be useful on a number of occasions later ("reduction homomorphisms" were also discussed at the end of Section 7.3 with reference to reducing the integers mod n).

Example

Let $R = \mathbb{Z}$ and consider the ideal $n\mathbb{Z}$ of $\mathbb{Z}$. Then the isomorphism above can be written

$$\mathbb{Z}[x]/n\mathbb{Z}[x] \cong \mathbb{Z}/n\mathbb{Z}[x]$$

and the natural projection map of $\mathbb{Z}[x]$ to $\mathbb{Z}/n\mathbb{Z}[x]$ by reducing the coefficients modulo n is a ring homomorphism. If n is composite, then the quotient ring is not an integral domain. If, however, n is a prime p, then $\mathbb{Z}/p\mathbb{Z}$ is a field and so $\mathbb{Z}/p\mathbb{Z}[x]$ is an integral domain (in fact, a Euclidean Domain, as we shall see shortly). We also see that the set of polynomials whose coefficients are divisible by p is a prime ideal in $\mathbb{Z}[x]$.

We close this section with a description of the natural extension to polynomial rings in *several* variables.

Definition. The *polynomial ring in the variables $x_1, x_2, \ldots, x_n$ with coefficients in R*, denoted $R[x_1, x_2, \ldots, x_n]$, is defined inductively by

$$R[x_1, x_2, \ldots, x_n] = R[x_1, x_2, \ldots, x_{n-1}][x_n]$$

This definition means that we can consider polynomials in n variables with coefficients in R simply as polynomials in *one* variable (say x_n) but now with coefficients that are themselves *polynomials in* $n-1$ *variables*. In a slightly more concrete formulation, a nonzero polynomial in $x_1, x_2, \dots, x_n$ with coefficients in R is a finite sum of nonzero *monomial terms*, i.e., a finite sum of elements of the form

$$ax_1^{d_1}x_2^{d_2}\dots x_n^{d_n}$$

where $a \in R$ (the *coefficient* of the term) and the d_i are nonnegative integers. A monic term $x_1^{d_1}x_2^{d_2}\dots x_n^{d_n}$ is called simply a *monomial* and is the *monomial part* of the term $ax_1^{d_1}x_2^{d_2}\dots x_n^{d_n}$. The exponent d_i is called the *degree in* x_i of the term and the sum

$$d = d_1 + d_2 + \cdots + d_n$$

is called the *degree* of the term. The ordered n-tuple $(d_1, d_2, \dots, d_n)$ is the *multidegree* of the term. The *degree* of a nonzero polynomial is the largest degree of any of its monomial terms. A polynomial is called *homogeneous* or a *form* if all its terms have the same degree. If f is a nonzero polynomial in n variables, the sum of all the monomial terms in f of degree k is called the *homogeneous component of* f *of degree* k. If f has degree d then f may be written uniquely as the sum $f_0 + f_1 + \cdots + f_d$ where f_k is the homogeneous component of f of degree k, for $0 \leq k \leq d$ (where some f_k may be zero).

Finally, to define a polynomial ring in an *arbitrary* number of variables with coefficients in R we take finite sums of monomial terms of the type above (but where the variables are not restricted to just $x_1, \dots, x_n$), with the natural addition and multiplication. Alternatively, we could define this ring as the *union* of *all* the polynomial rings in a *finite* number of the variables being considered.

Example

The polynomial ring $\mathbb{Z}[x, y]$ in two variables x and y with integer coefficients consists of all finite sums of monomial terms of the form ax^iy^j (of degree $i + j$). For example,

$$p(x, y) = 2x^3 + xy - y^2$$

and

$$q(x, y) = -3xy + 2y^2 + x^2y^3$$

are both elements of $\mathbb{Z}[x, y]$, of degrees 3 and 5, respectively. We have

$$p(x, y) + q(x, y) = 2x^3 - 2xy + y^2 + x^2y^3$$

and

$$p(x, y)q(x, y) = -6x^4y + 4x^3y^2 + 2x^5y^3 - 3x^2y^2 + 5xy^3 + x^3y^4 - 2y^4 - x^2y^5,$$

a polynomial of degree 8. To view this last polynomial, say, as a polynomial in y with coefficients in $\mathbb{Z}[x]$ as in the definition of several variable polynomial rings above, we would write the polynomial in the form

$$(-6x^4)y + (4x^3 - 3x^2)y^2 + (2x^5 + 5x)y^3 + (x^3 - 2)y^4 - (x^2)y^5.$$

The nonzero homogeneous components of $f = f(x, y) = p(x, y)q(x, y)$ are the polynomials $f_4 = -3x^2y^2 + 5xy^3 - 2y^4$ (degree 4), $f_5 = -6x^4y + 4x^3y^2$ (degree 5), $f_7 = x^3y^4 - x^2y^5$ (degree 7), and $f_8 = 2x^5y^3$ (degree 8).

Each of the statements in Proposition 1 is true for polynomial rings with an arbitrary number of variables. This follows by induction for finitely many variables and from the definition in terms of unions in the case of polynomial rings in arbitrarily many variables.

EXERCISES

1. Let $p(x, y, z) = 2x^2y - 3xy^3z + 4y^2z^5$ and $q(x, y, z) = 7x^2 + 5x^2y^3z^4 - 3x^2z^3$ be polynomials in $\mathbb{Z}[x, y, z]$.
   (a) Write each of p and q as a polynomial in x with coefficients in $\mathbb{Z}[y, z]$.
   (b) Find the degree of each of p and q.
   (c) Find the degree of p and q in each of the three variables x, y and z.
   (d) Compute pq and find the degree of pq in each of the three variables x, y and z.
   (e) Write pq as a polynomial in the variable z with coefficients in $\mathbb{Z}[x, y]$.
2. Repeat the preceding exercise under the assumption that the coefficients of p and q are in $\mathbb{Z}/3\mathbb{Z}$.
3. If R is a commutative ring and $x_1, x_2, \dots, x_n$ are independent variables over R, prove that $R[x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(n)}]$ is isomorphic to $R[x_1, x_2, \dots, x_n]$ for any permutation π of $\{1, 2, \dots, n\}$.
4. Prove that the ideals (x) and (x, y) are prime ideals in $\mathbb{Q}[x, y]$ but only the latter ideal is a maximal ideal.
5. Prove that (x, y) and $(2, x, y)$ are prime ideals in $\mathbb{Z}[x, y]$ but only the latter ideal is a maximal ideal.
6. Prove that (x, y) is not a principal ideal in $\mathbb{Q}[x, y]$.
7. Let R be a commutative ring with 1. Prove that a polynomial ring in more than one variable over R is not a Principal Ideal Domain.
8. Let F be a field and let $R = F[x, x^2y, x^3y^2, \dots, x^ny^{n-1}, \dots]$ be a subring of the polynomial ring $F[x, y]$.
   (a) Prove that the fields of fractions of R and $F[x, y]$ are the same.
   (b) Prove that R contains an ideal that is not finitely generated.
9. Prove that a polynomial ring in infinitely many variables with coefficients in any commutative ring contains ideals that are not finitely generated.
10. Prove that the ring $\mathbb{Z}[x_1, x_2, x_3, \dots]/(x_1x_2, x_3x_4, x_5x_6, \dots)$ contains infinitely many minimal prime ideals (cf. Exercise 36 of Section 7.4).
11. Show that the radical of the ideal $I = (x, y^2)$ in $\mathbb{Q}[x, y]$ is (x, y) (cf. Exercise 30, Section 7.4). Deduce that I is a primary ideal that is not a power of a prime ideal (cf. Exercise 41, Section 7.4).
12. Let $R = \mathbb{Q}[x, y, z]$ and let bars denote passage to $\mathbb{Q}[x, y, z]/(xy - z^2)$. Prove that $\overline{P} = (\overline{x}, \overline{z})$ is a prime ideal. Show that $\overline{xy} \in \overline{P}^2$ but that no power of $\overline{y}$ lies in $\overline{P}^2$. (This shows $\overline{P}$ is a prime ideal whose square is *not* a primary ideal — cf. Exercise 41, Section 7.4).
13. Prove that the rings $F[x, y]/(y^2 - x)$ and $F[x, y]/(y^2 - x^2)$ are not isomorphic for any field F.
14. Let R be an integral domain and let i, j be relatively prime integers. Prove that the ideal $(x^i - y^j)$ is a prime ideal in $R[x, y]$. [Consider the ring homomorphism φ from $R[x, y]$ to $R[t]$ defined by mapping x to t^j and mapping y to t^i. Show that an element of $R[x, y]$

differs from an element in $(x^i - y^j)$ by a polynomial $f(x)$ of degree at most $j-1$ in y and observe that the exponents of $\varphi(x^r y^s)$ are distinct for $0 \leq s < j$.]

15. Let $p(x_1, x_2, \ldots, x_n)$ be a homogeneous polynomial of degree k in $R[x_1, \ldots, x_n]$. Prove that for all $\lambda \in R$ we have $p(\lambda x_1, \lambda x_2, \ldots, \lambda x_n) = \lambda^k p(x_1, x_2, \ldots, x_n)$.

16. Prove that the product of two homogeneous polynomials is again homogeneous.

17. An ideal I in $R[x_1, \ldots, x_n]$ is called a *homogeneous ideal* if whenever $p \in I$ then each homogeneous component of p is also in I. Prove that an ideal is a homogeneous ideal if and only if it may be generated by homogeneous polynomials. [Use induction on degrees to show the "if" implication.]

The following exercise shows that some care must be taken when working with polynomials over noncommutative rings R (the ring operations in $R[x]$ are defined in the same way as for commutative rings R), in particular when considering polynomials as functions.

18. Let R be an arbitrary ring and let Func(R) be the ring of all functions from R to itself. If $p(x) \in R[x]$ is a polynomial, let $f_p \in$ Func(R) be the function on R defined by $f_p(r) = p(r)$ (the usual way of viewing a polynomial in $R[x]$ as defining a function on R by "evaluating at r").

  (a) For fixed $a \in R$, prove that "evaluation at a" is a ring homomorphism from Func(R) to R (cf. Example 4 following Theorem 7 in Section 7.3).

  (b) Prove that the map $\varphi : R[x] \to$ Func(R) defined by $\varphi(p(x)) = f_p$ is not a ring homomorphism in general. Deduce that polynomial identities need not give corresponding identities when the polynomials are viewed as functions. [If $R = \mathbb{H}$ is the ring of real Hamilton Quaternions show that $p(x) = x^2 + 1$ factors as $(x+i)(x-i)$, but that $p(j) = 0$ while $(j+i)(j-i) \neq 0$.]

  (c) For fixed $a \in R$, prove that the composite "evaluation at a" of the maps in (a) and (b) mapping $R[x]$ to R is a ring homomorphism if and only if a is in the center of R.

9.2 POLYNOMIAL RINGS OVER FIELDS I

We now consider more carefully the situation where the coefficient ring is a *field* F. We can define a *norm* on $F[x]$ by defining $N(p(x)) =$ degree of $p(x)$ (where we set $N(0) = 0$). From elementary algebra we know that we can divide one polynomial with, say, rational coefficients by another (nonzero) polynomial with rational coefficients to obtain a quotient and remainder. The same is true over any field.

Theorem 3. Let F be a field. The polynomial ring $F[x]$ is a Euclidean Domain. Specifically, if $a(x)$ and $b(x)$ are two polynomials in $F[x]$ with $b(x)$ nonzero, then there are *unique* $q(x)$ and $r(x)$ in $F[x]$ such that

$$a(x) = q(x)b(x) + r(x) \qquad \text{with } r(x) = 0 \text{ or degree } r(x) < \text{degree } b(x)\,.$$

Proof: If $a(x)$ is the zero polynomial then take $q(x) = r(x) = 0$. We may therefore assume $a(x) \neq 0$ and prove the existence of $q(x)$ and $r(x)$ by induction on $n =$ degree $a(x)$. Let $b(x)$ have degree m. If $n < m$ take $q(x) = 0$ and $r(x) = a(x)$. Otherwise $n \geq m$. Write

$$a(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

and

$$b(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0.$$

Then the polynomial $a'(x) = a(x) - \frac{a_n}{b_m}x^{n-m}b(x)$ is of degree less than n (we have arranged to subtract the leading term from $a(x)$). Note that this polynomial is well defined because the coefficients are taken from a *field* and $b_m \neq 0$. By induction then, there exist polynomials $q'(x)$ and $r(x)$ with

$$a'(x) = q'(x)b(x) + r(x) \qquad \text{with } r(x) = 0 \text{ or degree } r(x) < \text{degree } b(x).$$

Then, letting $q(x) = q'(x) + \frac{a_n}{b_m}x^{n-m}$ we have

$$a(x) = q(x)b(x) + r(x) \qquad \text{with } r(x) = 0 \text{ or degree } r(x) < \text{degree } b(x)$$

completing the induction step.

As for the uniqueness, suppose $q_1(x)$ and $r_1(x)$ also satisfied the conditions of the theorem. Then both $a(x) - q(x)b(x)$ and $a(x) - q_1(x)b(x)$ are of degree less than m = degree $b(x)$. The difference of these two polynomials, i.e., $b(x)(q(x) - q_1(x))$ is also of degree less than m. But the degree of the product of two nonzero polynomials is the sum of their degrees (since F is an integral domain), hence $q(x) - q_1(x)$ must be 0, that is, $q(x) = q_1(x)$. This implies $r(x) = r_1(x)$, completing the proof.

Corollary 4. If F is a field, then $F[x]$ is a Principal Ideal Domain and a Unique Factorization Domain.

Proof: This is immediate from the results of the last chapter.

Recall also from Corollary 8 in Section 8.2 that if R is any commutative ring such that $R[x]$ is a Principal Ideal Domain (or Euclidean Domain) then R must be a field. We shall see in the next section, however, that $R[x]$ is a Unique Factorization Domain whenever R itself is a Unique Factorization Domain.

Examples

(1) By the above remarks the ring $\mathbb{Z}[x]$ is not a Principal Ideal Domain. As we have already seen (Example 3 beginning of Section 7.4) the ideal $(2, x)$ is not principal in this ring.

(2) $\mathbb{Q}[x]$ is a Principal Ideal Domain since the coefficients lie in the field $\mathbb{Q}$. The ideal generated in $\mathbb{Z}[x]$ by 2 and x is not principal in the subring $\mathbb{Z}[x]$ of $\mathbb{Q}[x]$. However, the ideal generated in $\mathbb{Q}[x]$ is principal; in fact it is the entire ring (so has 1 as a generator) since 2 is a unit in $\mathbb{Q}[x]$.

(3) If p is a prime, the ring $\mathbb{Z}/p\mathbb{Z}[x]$ obtained by reducing $\mathbb{Z}[x]$ modulo the prime ideal (p) is a Principal Ideal Domain, since the coefficients lie in the field $\mathbb{Z}/p\mathbb{Z}$. This example shows that the quotient of a ring which is not a Principal Ideal Domain *may* be a Principal Ideal Domain. To follow the ideal $(2, x)$ above in this example, note that if $p = 2$, then the ideal $(2, x)$ reduces to the ideal (x) in the quotient $\mathbb{Z}/2\mathbb{Z}[x]$, which is a proper (maximal) ideal. If $p \neq 2$, then 2 is a unit in the quotient, so the ideal $(2, x)$ reduces to the entire ring $\mathbb{Z}/p\mathbb{Z}[x]$.

(4) $\mathbb{Q}[x, y]$, the ring of polynomials in two variables with rational coefficients, is *not* a Principal Ideal Domain since this ring is $\mathbb{Q}[x][y]$ and $\mathbb{Q}[x]$ is not a field (any element of positive degree is not invertible). It is an exercise to see that the ideal (x, y) is not a principal ideal in this ring. We shall see shortly that $\mathbb{Q}[x, y]$ *is* a Unique Factorization Domain.

We note that the quotient and remainder in the Division Algorithm applied to $a(x), b(x) \in F[x]$ are *independent of field extensions* in the following sense. Suppose the field F is contained in the field E and $a(x) = Q(x)b(x) + R(x)$ for some $Q(x)$, $R(x)$ satisfying the conditions of Theorem 3 in $E[x]$. Write $a(x) = q(x)b(x) + r(x)$ for some $q(x), r(x) \in F[x]$ and apply the uniqueness condition of Theorem 3 in the ring $E[x]$ to deduce that $Q(x) = q(x)$ and $R(x) = r(x)$. In particular, $b(x)$ divides $a(x)$ in the ring $E[x]$ if and only if $b(x)$ divides $a(x)$ in $F[x]$. Also, the greatest common divisor of $a(x)$ and $b(x)$ (which can be obtained from the Euclidean Algorithm) is the same, once we make it unique by specifying it to be monic, whether these elements are viewed in $F[x]$ or in $E[x]$.

EXERCISES

Let F be a field and let x be an indeterminate over F.

1. Let $f(x) \in F[x]$ be a polynomial of degree $n \geq 1$ and let bars denote passage to the quotient $F[x]/(f(x))$. Prove that for each $\overline{g(x)}$ there is a unique polynomial $g_0(x)$ of degree $\leq n - 1$ such that $\overline{g(x)} = \overline{g_0(x)}$ (equivalently, the elements $\bar{1}, \bar{x}, \ldots, \overline{x^{n-1}}$ are a *basis* of the vector space $F[x]/(f(x))$ over F — in particular, the dimension of this space is n). [Use the Division Algorithm.]

2. Let F be a finite field of order q and let $f(x)$ be a polynomial in $F[x]$ of degree $n \geq 1$. Prove that $F[x]/(f(x))$ has q^n elements. [Use the preceding exercise.]

3. Let $f(x)$ be a polynomial in $F[x]$. Prove that $F[x]/(f(x))$ is a field if and only if $f(x)$ is irreducible. [Use Proposition 7, Section 8.2.]

4. Let F be a finite field. Prove that $F[x]$ contains infinitely many primes. (Note that over an infinite field the polynomials of degree 1 are an infinite set of primes in the ring of polynomials).

5. Exhibit *all* the ideals in the ring $F[x]/(p(x))$, where F is a field and $p(x)$ is a polynomial in $F[x]$ (describe them in terms of the factorization of $p(x)$).

6. Describe (briefly) the ring structure of the following rings:
(a) $\mathbb{Z}[x]/(2)$, **(b)** $\mathbb{Z}[x]/(x)$, **(c)** $\mathbb{Z}[x]/(x^2)$, **(d)** $\mathbb{Z}[x, y]/(x^2, y^2, 2)$.
Show that $\alpha^2 = 0$ or 1 for every α in the last ring and determine those elements with $\alpha^2 = 0$. Determine the characteristics of each of these rings (cf. Exercise 26, Section 7.3).

7. Determine all the ideals of the ring $\mathbb{Z}[x]/(2, x^3 + 1)$.

8. Determine the greatest common divisor of $a(x) = x^3 - 2$ and $b(x) = x + 1$ in $\mathbb{Q}[x]$ and write it as a linear combination (in $\mathbb{Q}[x]$) of $a(x)$ and $b(x)$.

9. Determine the greatest common divisor of $a(x) = x^5 + 2x^3 + x^2 + x + 1$ and the polynomial $b(x) = x^5 + x^4 + 2x^3 + 2x^2 + 2x + 1$ in $\mathbb{Q}[x]$ and write it as a linear combination (in $\mathbb{Q}[x]$) of $a(x)$ and $b(x)$.

10. Determine the greatest common divisor of $a(x) = x^3 + 4x^2 + x - 6$ and $b(x) = x^5 - 6x + 5$ in $\mathbb{Q}[x]$ and write it as a linear combination (in $\mathbb{Q}[x]$) of $a(x)$ and $b(x)$.

11. Suppose $f(x)$ and $g(x)$ are two nonzero polynomials in $\mathbb{Q}[x]$ with greatest common divisor $d(x)$.

(a) Given $h(x) \in \mathbb{Q}[x]$, show that there are polynomials $a(x), b(x) \in \mathbb{Q}[x]$ satisfying the equation $a(x)f(x) + b(x)g(x) = h(x)$ if and only if $h(x)$ is divisible by $d(x)$.

(b) If $a_0(x), b_0(x) \in \mathbb{Q}[x]$ are particular solutions to the equation in (a), show that the full set of solutions to this equation is given by

$$a(x) = a_0(x) + m(x)\frac{g(x)}{(x)d}$$
$$b(x) = b_0(x) - m(x)\frac{f(x)}{d(x)}$$

as $m(x)$ ranges over the polynomials in $\mathbb{Q}[x]$. [cf. Exercise 4 in Section 8.1]

12. Let $F[x, y_1, y_2, \ldots]$ be the polynomial ring in the infinite set of variables $x, y_1, y_2, \ldots$ over the field F, and let I be the ideal $(x - y_1^2, y_1 - y_2^2, \ldots, y_i - y_{i+1}^2, \ldots)$ in this ring. Define R to be the ring $F[x, y_1, y_2, \ldots]/I$, so that in R the square of each y_{i+1} is y_i and $y_1^2 = x$ modulo I, i.e., x has a 2^ith root, for every i. Denote the image of y_i in R as $x^{1/2^i}$. Let R_n be the subring of R generated by F and $x^{1/2^n}$.

(a) Prove that $R_1 \subseteq R_2 \subseteq \cdots$ and that R is the union of all R_n, i.e., $R = \cup_{n=1}^{\infty} R_n$.

(b) Prove that R_n is isomorphic to a polynomial ring in one variable over F, so that R_n is a P.I.D. Deduce that R is a Bezout Domain (cf. Exercise 7 in Section 8.2). [First show that the ring $S_n = F[x, y_1, \ldots, y_n]/(x - y_1^2, y_1 - y_2^2, \ldots, y_{n-1} - y_n^2)$ is isomorphic to the polynomial ring $F[y_n]$. Then show any polynomial relation y_n satisfies in R_n gives a corresponding relation in S_N for some $N \geq n$.]

(c) Prove that the ideal generated by $x, x^{1/2}, x^{1/4}, \ldots$ in R is not finitely generated (so R is not a P.I.D.).

13. This exercise introduces a noncommutative ring which is a "right" Euclidean Domain (and a "left" Principal Ideal Domain) but is not a "left" Euclidean Domain (and not a "right" Principal Ideal Domain). Let F be a field of characteristic p in which not every element is a p^{th} power: $F \neq F^p$ (for example the field $F = \mathbb{F}_p(t)$ of rational functions in the variable t with coefficients in $\mathbb{F}_p$ is such a field). Let $R = F\{x\}$ be the "twisted" polynomial ring of polynomials $\sum_{i=0}^{n} a_i x^i$ in x with coefficients in F with the usual (termwise) addition

$$\sum_{i=0}^{n} a_i x^i + \sum_{i=0}^{n} b_i x^i = \sum_{i=0}^{n} (a_i + b_i) x^i$$

but with a noncommutative multiplication defined by

$$\left(\sum_{i=0}^{n} a_i x^i\right)\left(\sum_{j=0}^{m} b_j x^j\right) = \sum_{k=0}^{n+m}\left(\sum_{i+j=k} a_i b_j^{p^i}\right) x^k .$$

This multiplication arises from defining $xa = a^p x$ for every $a \in F$ (so the powers of x do not commute with the coefficients) and extending in a natural way. Let N be the norm defined by taking the degree of a polynomial in R: $N(f) = \deg(f)$.

(a) Show that $x^k a = a^{p^k} x^k$ for every $a \in F$ and every integer $k \geq 0$ and that R is a ring with this definition of multiplication. [Use the fact that $(a + b)^p = a^p + b^p$ for every $a, b \in F$ since F has characteristic p, so also $(a + b)^{p^k} = a^{p^k} + b^{p^k}$ for every $a, b \in F$.]

(b) Prove that the degree of a product of two elements of R is the sum of the degrees of the elements. Prove that R has no zero divisors.

(c) Prove that R is "right Euclidean" with respect to N, i.e., for any polynomials $f, g \in R$ with $g \neq 0$, there exist polynomials q and r in R with

$$f = qg + r \qquad \text{with } r = 0 \text{ or } \deg(r) < \deg(g).$$

Use this to prove that every *left* ideal of R is principal.

(d) Let $f = \theta x$ for some $\theta \in F$, $\theta \notin F^p$ and let $g = x$. Prove that there are no polynomials q and r in R with

$$f = gq + r \qquad \text{with } r = 0 \text{ or } \deg(r) < \deg(g),$$

so in particular R is not "left Euclidean" with respect to N. Prove that the right ideal of R generated by x and θx is not principal. Conclude that R is not "left Euclidean" with respect to *any* norm.

9.3 POLYNOMIAL RINGS THAT ARE UNIQUE FACTORIZATION DOMAINS

We have seen in Proposition 1 that if R is an integral domain then $R[x]$ is also an integral domain. Also, such an R can be embedded in its field of fractions F (Theorem 15, Section 7.5), so that $R[x] \subseteq F[x]$ is a subring, and $F[x]$ is a Euclidean Domain (hence a Principal Ideal Domain and a Unique Factorization Domain). Many computations for $R[x]$ may be accomplished in $F[x]$ at the expense of allowing fractional coefficients. This raises the immediate question of how computations (such as factorizations of polynomials) in $F[x]$ can be used to give information in $R[x]$.

For instance, suppose $p(x)$ is a polynomial in $R[x]$. Since $F[x]$ is a Unique Factorization Domain we can factor $p(x)$ uniquely into a product of irreducibles in $F[x]$. It is natural to ask whether we can do the same in $R[x]$, i.e., is $R[x]$ a Unique Factorization Domain? In general the answer is no because if $R[x]$ were a Unique Factorization Domain, the constant polynomials would have to be uniquely factored into irreducible elements of $R[x]$, necessarily of degree 0 since the degrees of products add, that is, R would itself have to be a Unique Factorization Domain. Thus if R is an integral domain which is not a Unique Factorization Domain, $R[x]$ cannot be a Unique Factorization Domain. On the other hand, it turns out that if R is a Unique Factorization Domain, then $R[x]$ is also a Unique Factorization Domain. The method of proving this is to first factor uniquely in $F[x]$ and then "clear denominators" to obtain a unique factorization in $R[x]$. The first step in making this precise is to compare the factorization of a polynomial in $F[x]$ to a factorization in $R[x]$.

Proposition 5. *(Gauss' Lemma)* Let R be a Unique Factorization Domain with field of fractions F and let $p(x) \in R[x]$. If $p(x)$ is reducible in $F[x]$ then $p(x)$ is reducible in $R[x]$. More precisely, if $p(x) = A(x)B(x)$ for some nonconstant polynomials $A(x), B(x) \in F[x]$, then there are nonzero elements $r, s \in F$ such that $rA(x) = a(x)$ and $sB(x) = b(x)$ both lie in $R[x]$ and $p(x) = a(x)b(x)$ is a factorization in $R[x]$.

Proof: The coefficients of the polynomials on the right hand side of the equation $p(x) = A(x)B(x)$ are elements in the field F, hence are quotients of elements from the Unique Factorization Domain R. Multiplying through by a common denominator for all these coefficients, we obtain an equation $dp(x) = a'(x)b'(x)$ where now $a'(x)$ and $b'(x)$ are elements of $R[x]$ and d is a nonzero element of R. If d is a unit in R, the proposition is true with $a(x) = d^{-1}a'(x)$ and $b(x) = b'(x)$. Assume d is not a unit and write d as a product of irreducibles in R, say $d = p_1 \cdots p_n$. Since p_1 is irreducible in R, the ideal (p_1) is prime (cf. Proposition 12, Section 8.3), so by Proposition 2 above, the ideal $p_1R[x]$ is prime in $R[x]$ and $(R/p_1R)[x]$ is an integral domain. Reducing the equation $dp(x) = a'(x)b'(x)$ modulo p_1, we obtain the equation $0 = \overline{a'(x)}\,\overline{b'(x)}$ in this integral domain (the bars denote the images of these polynomials in the quotient ring), hence one of the two factors, say $\overline{a'(x)}$ must be 0. But this means all the coefficients of $a'(x)$ are divisible by p_1, so that $\frac{1}{p_1}a'(x)$ also has coefficients in R. In other words, in the equation $dp(x) = a'(x)b'(x)$ we can cancel a factor of p_1 from d (on the left) and from either $a'(x)$ or $b'(x)$ (on the right) and still have an equation in $R[x]$. But now the factor d on the left hand side has one fewer irreducible factors. Proceeding in the same fashion with each of the remaining factors of d, we can cancel all of the factors of d into the two polynomials on the right hand side, leaving an equation $p(x) = a(x)b(x)$ with $a(x), b(x) \in R[x]$ and with $a(x), b(x)$ being F-multiples of $A(x)$, $B(x)$, respectively. This completes the proof.

Note that we cannot prove that $a(x)$ and $b(x)$ are necessarily R-multiples of $A(x)$, $B(x)$, respectively, because, for example, we could factor x^2 in $\mathbb{Q}[x]$ with $A(x) = 2x$ and $B(x) = \frac{1}{2}x$ but no *integer* multiples of $A(x)$ and $B(x)$ give a factorization of x^2 in $\mathbb{Z}[x]$.

The elements of the ring R become *units* in the Unique Factorization Domain $F[x]$ (the units in $F[x]$ being the nonzero elements of F). For example, $7x$ factors in $\mathbb{Z}[x]$ into a product of two irreducibles: 7 and x (so $7x$ is not irreducible in $\mathbb{Z}[x]$), whereas $7x$ is the unit 7 times the irreducible x in $\mathbb{Q}[x]$ (so $7x$ is irreducible in $\mathbb{Q}[x]$). The following corollary shows that this is essentially the *only* difference between the irreducible elements in $R[x]$ and those in $F[x]$.

Corollary 6. Let R be a Unique Factorization Domain, let F be its field of fractions and let $p(x) \in R[x]$. Suppose the greatest common divisor of the coefficients of $p(x)$ is 1. Then $p(x)$ is irreducible in $R[x]$ if and only if it is irreducible in $F[x]$. In particular, if $p(x)$ is a monic polynomial that is irreducible in $R[x]$, then $p(x)$ is irreducible in $F[x]$.

Proof: By Gauss' Lemma above, if $p(x)$ is reducible in $F[x]$, then it is reducible in $R[x]$. Conversely, the assumption on the greatest common divisor of the coefficients of $p(x)$ implies that if it is reducible in $R[x]$, then $p(x) = a(x)b(x)$ where neither $a(x)$ nor $b(x)$ are constant polynomials in $R[x]$. This same factorization shows that $p(x)$ is reducible in $F[x]$, completing the proof.

Theorem 7. R is a Unique Factorization Domain if and only if $R[x]$ is a Unique Factorization Domain.

Proof: We have indicated above that $R[x]$ a Unique Factorization Domain forces R to be a Unique Factorization Domain. Suppose conversely that R is a Unique Factorization Domain, F is its field of fractions and $p(x)$ is a nonzero element of $R[x]$. Let d be the greatest common divisor of the coefficients of $p(x)$, so that $p(x) = dp'(x)$, where the g.c.d. of the coefficients of $p'(x)$ is 1. Such a factorization of $p(x)$ is unique up to a change in d (so up to a unit in R), and since d can be factored uniquely into irreducibles in R (and these are also irreducibles in the larger ring $R[x]$), it suffices to prove that $p'(x)$ can be factored uniquely into irreducibles in $R[x]$. Thus we may assume that the greatest common divisor of the coefficients of $p(x)$ is 1. We may further assume $p(x)$ is not a unit in $R[x]$, i.e., degree $p(x) > 0$.

Since $F[x]$ is a Unique Factorization Domain, $p(x)$ can be factored uniquely into irreducibles in $F[x]$. By Gauss' Lemma, such a factorization implies there is a factorization of $p(x)$ in $R[x]$ whose factors are F-multiples of the factors in $F[x]$. Since the greatest common divisor of the coefficients of $p(x)$ is 1, the g.c.d. of the coefficients in each of these factors in $R[x]$ must be 1. By Corollary 6, each of these factors is an irreducible in $R[x]$. This shows that $p(x)$ can be written as a finite product of irreducibles in $R[x]$.

The uniqueness of the factorization of $p(x)$ follows from the uniqueness in $F[x]$. Suppose

$$p(x) = q_1(x) \cdots q_r(x) = q_1'(x) \cdots q_s'(x)$$

are two factorizations of $p(x)$ into irreducibles in $R[x]$. Since the g.c.d. of the coefficients of $p(x)$ is 1, the same is true for each of the irreducible factors above — in particular, each has positive degree. By Corollary 6, each $q_i(x)$ and $q_j'(x)$ is an irreducible in $F[x]$. By unique factorization in $F[x]$, $r = s$ and, possibly after rearrangement, $q_i(x)$ and $q_i'(x)$ are associates in $F[x]$ for all $i \in \{1, \dots, r\}$. It remains to show they are associates in $R[x]$. Since the units of $F[x]$ are precisely the elements of $F^\times$ we need to consider when $q(x) = \frac{a}{b}q'(x)$ for some $q(x), q'(x) \in R[x]$ and nonzero elements a, b of R, where the greatest common divisor of the coefficients of each of $q(x)$ and $q'(x)$ is 1. In this case $bq(x) = aq'(x)$; the g.c.d. of the coefficients on the left hand side is b and on the right hand side is a. Since in a Unique Factorization Domain the g.c.d. of the coefficients of a nonzero polynomial is unique up to units, $a = ub$ for some unit u in R. Thus $q(x) = uq'(x)$ and so $q(x)$ and $q'(x)$ are associates in R as well. This completes the proof.

Corollary 8. If R is a Unique Factorization Domain, then a polynomial ring in an arbitrary number of variables with coefficients in R is also a Unique Factorization Domain.

Proof: For finitely many variables, this follows by induction from Theorem 7, since a polynomial ring in n variables can be considered as a polynomial ring in one variable with coefficients in a polynomial ring in $n-1$ variables. The general case follows from the definition of a polynomial ring in an arbitrary number of variables as the union of polynomial rings in finitely many variables.

Examples

(1) $\mathbb{Z}[x]$, $\mathbb{Z}[x, y]$, etc. are Unique Factorization Domains. The ring $\mathbb{Z}[x]$ gives an example of a Unique Factorization Domain that is not a Principal Ideal Domain.

(2) Similarly, $\mathbb{Q}[x]$, $\mathbb{Q}[x, y]$, etc. are Unique Factorization Domains.

We saw earlier that if R is a Unique Factorization Domain with field of fractions F and $p(x) \in R[x]$, then we can factor out the greatest common divisor d of the coefficients of $p(x)$ to obtain $p(x) = dp'(x)$, where $p'(x)$ is irreducible in both $R[x]$ and $F[x]$. Suppose now that R is an *arbitrary* integral domain with field of fractions F. In R the notion of greatest common divisor may not make sense, however one might still ask if, say, a *monic* polynomial which is irreducible in $R[x]$ is still irreducible in $F[x]$ (i.e., whether the last statement in Corollary 6 is true).

Note first that if a monic polynomial $p(x)$ is reducible, it must have a factorization $p(x) = a(x)b(x)$ in $R[x]$ with both $a(x)$ and $b(x)$ *monic, nonconstant* polynomials (recall that the leading term of $p(x)$ is the product of the leading terms of the factors, so the leading coefficients of both $a(x)$ and $b(x)$ are units — we can thus arrange these to be 1). In other words, a nonconstant *monic* polynomial $p(x)$ is irreducible if and only if it cannot be factored as a product of two *monic* polynomials of smaller degree.

We now see that it is not true that if R is an arbitrary integral domain and $p(x)$ is a monic irreducible polynomial in $R[x]$, then $p(x)$ is irreducible in $F[x]$. For example, let $R = \mathbb{Z}[2i] = \{a + 2bi \mid a, b \in \mathbb{Z}\}$ (a subring of the complex numbers) and let $p(x) = x^2+1$. Then the fraction field of R is $F = \{a+bi \mid a, b \in \mathbb{Q}\}$. The polynomial $p(x)$ factors uniquely into a product of two linear factors in $F[x]$: $x^2+1 = (x-i)(x+i)$ so in particular, *$p(x)$ is reducible in $F[x]$*. Neither of these factors lies in $R[x]$ (because $i \notin R$) so *$p(x)$ is irreducible in $R[x]$*. In particular, by Corollary 6, *$\mathbb{Z}[2i]$ is not a Unique Factorization Domain.*

EXERCISES

1. Let R be an integral domain with quotient field F and let $p(x)$ be a monic polynomial in $R[x]$. Assume that $p(x) = a(x)b(x)$ where $a(x)$ and $b(x)$ are monic polynomials in $F[x]$ of smaller degree than $p(x)$. Prove that if $a(x) \notin R[x]$ then R is not a Unique Factorization Domain. Deduce that $\mathbb{Z}[2\sqrt{2}]$ is not a U.F.D.
2. Prove that if $f(x)$ and $g(x)$ are polynomials with rational coefficients whose product $f(x)g(x)$ has integer coefficients, then the product of any coefficient of $g(x)$ with any coefficient of $f(x)$ is an integer.
3. Let F be a field. Prove that the set R of polynomials in $F[x]$ whose coefficient of x is equal to 0 is a subring of $F[x]$ and that R is not a U.F.D. [Show that $x^6 = (x^2)^3 = (x^3)^2$ gives two distinct factorizations of x^6 into irreducibles.]
4. Let $R = \mathbb{Z}+x\mathbb{Q}[x] \subset \mathbb{Q}[x]$ be the set of polynomials in x with rational coefficients whose constant term is an integer.
   (a) Prove that R is an integral domain and its units are ± 1.
   (b) Show that the irreducibles in R are $\pm p$ where p is a prime in $\mathbb{Z}$ and the polynomials $f(x)$ that are irreducible in $\mathbb{Q}[x]$ and have constant term ± 1. Prove that these irreducibles are prime in R.
   (c) Show that x cannot be written as the product of irreducibles in R (in particular, x is not irreducible) and conclude that R is not a U.F.D.
   (d) Show that x is not a prime in R and describe the quotient ring $R/(x)$.
5. Let $R = \mathbb{Z}+x\mathbb{Q}[x] \subset \mathbb{Q}[x]$ be the ring considered in the previous exercise.
   (a) Suppose that $f(x), g(x) \in \mathbb{Q}[x]$ are two nonzero polynomials with rational coefficients and that x^r is the largest power of x dividing both $f(x)$ and $g(x)$ in $\mathbb{Q}[x]$, (i.e., r is the degree of the lowest order term appearing in either $f(x)$ or $g(x)$). Let f_r and